

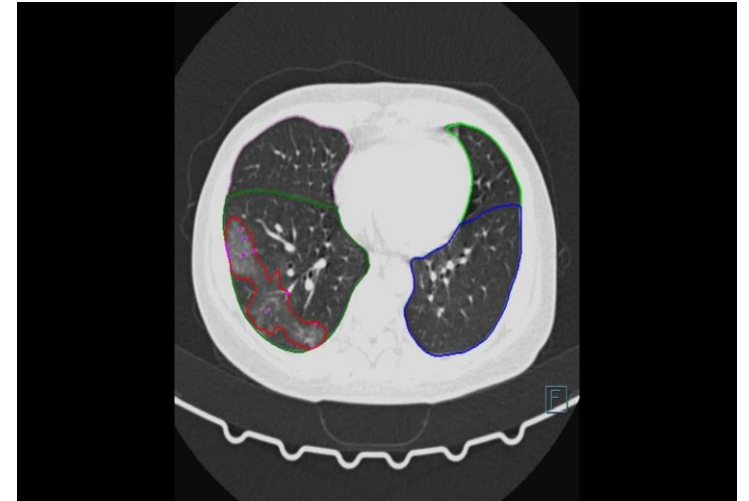
Artificial Intelligence as a driver for Innovation in Digital Health

Francesco Pugliese, Central Directorate for Methodology and Design of Statistical Processes, **Istat**

Roma | 18 May 2026 | La Sapienza

AI in Digital Health

- **AI in digital health** has enormously diverse applications, ranging from prevention to hospital management. Some are already mature and used every day, while others are still experimental but highly promising.
- The **best-known** use is diagnostic support:
 - automatic analysis of **X-rays, CT scans, MRIs, and mammograms**
 - early detection of tumors or lesions
 - identification of anomalies in ECGs and EEGs
 - systems that suggest probable diagnoses to doctors



AI in Digital Health

- **Personalized Medicine:** AI can combine:
 - clinical data
 - genomics
 - lifestyle
 - wearable devicesto create **personalized therapies**.

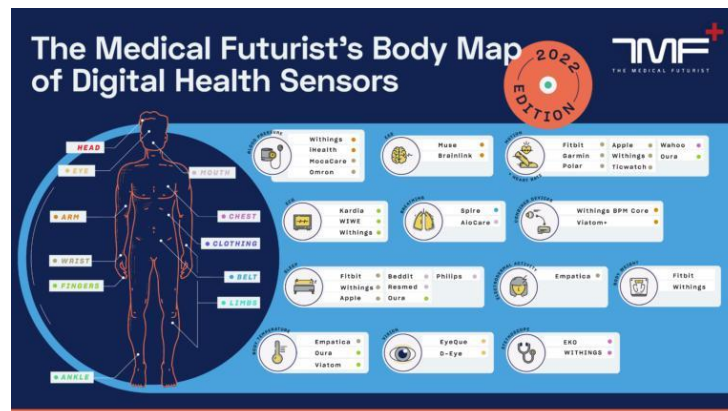
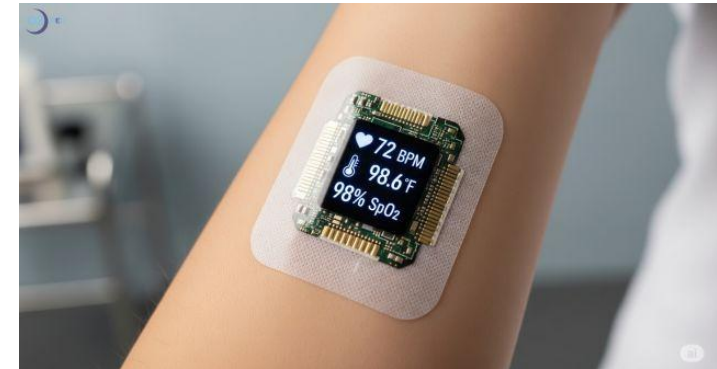
Examples:

- 1) predicting which **cancer** therapy will work best
- 2) optimized **drug** dosage
- 3) identification of high-risk patients
- 4) Fields such as **Precision Medicine** and **Bioinformatics** play a major role here.



Remote Monitoring and Wearables

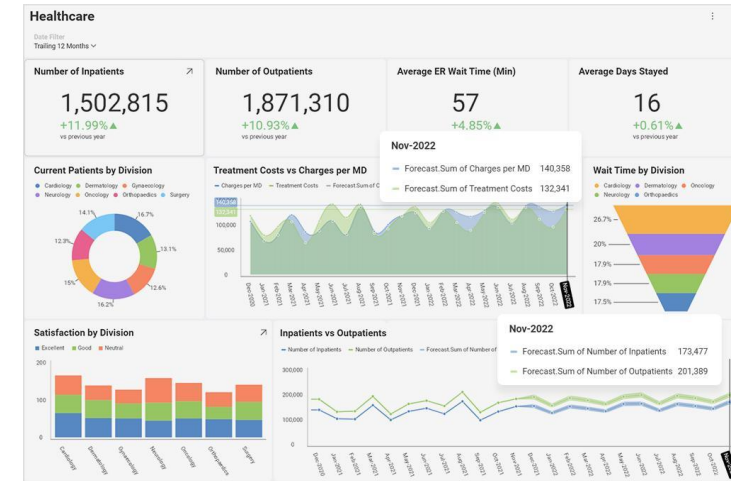
- **Smartwatches and sensors** can use **AI** to:
 - detect arrhythmias
 - monitor blood glucose levels
 - predict epileptic seizures
 - monitor cardiac parameters
 - identify falls in elderly people
- Products such as Apple Watch and Fitbit already integrate AI algorithms.



Healthcare Chatbots and Virtual Assistants

- Very widespread applications include:

- automated triage
- appointment booking
- healthcare FAQs
- basic psychological support
- therapy reminders



Many **hospitals** are experimenting with chatbots based on LLMs to:

- **summarize** medical records
- **assist** healthcare professionals
- **automate** documentation

Large Language Models (LLMs) are becoming increasingly important in healthcare digital transformation.

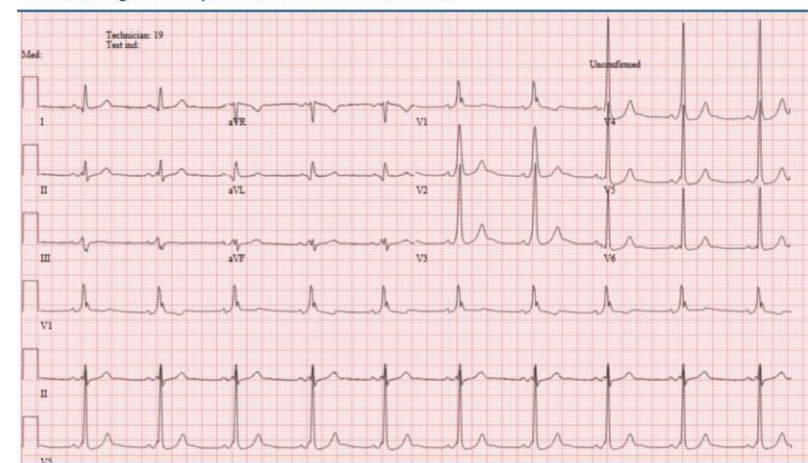
Public Health and Statistics

- Also **very interesting** for statistical agencies and institutions:
 - epidemiological surveillance
 - healthcare sentiment analysis
 - epidemic spread prediction
 - automatic classification of healthcare data
 - management of healthcare contact centers

Here, AI and Official Statistics are becoming increasingly interconnected.

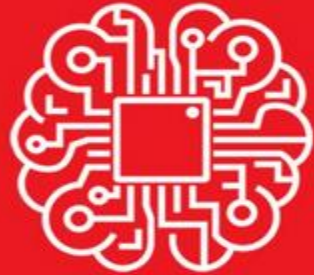


Eletrcardiograma compatível com Wolf Parkinson White.



Guia TdC®

Difference between AI and ML



ARTIFICIAL
INTELLIGENCE

—
If it is written
in **PowerPoint**,
It's probably
**Artificial
Intelligence**



MACHINE
LEARNING

—
If it is written
in **Python**;
It's probably
**Machine
Learning**

MACHINE LEARNING ENGINEER



What My Parent Think I do



What My Friend Think I do



What Society Think I do



What Media Think I do



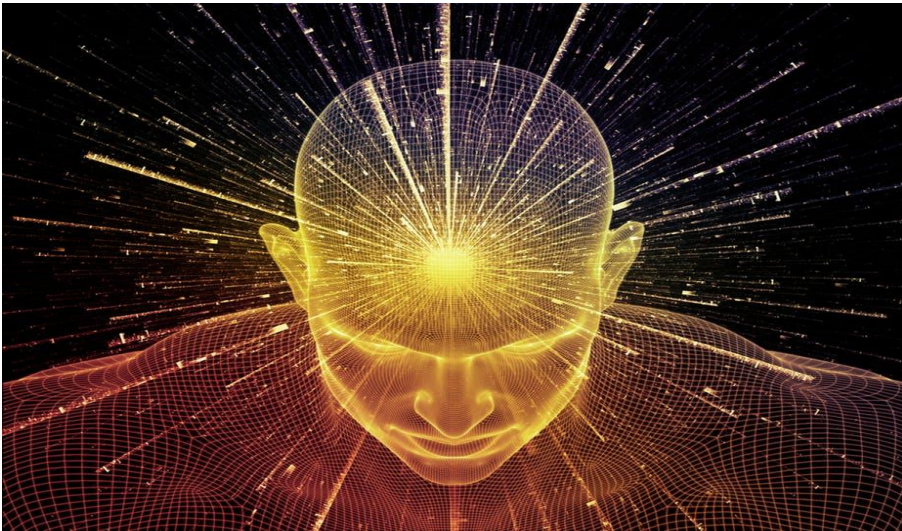
What I Think I do

```
1 import tensorflow as tf
2 import torch
3
```

What I really do

WHAT IS CONSCIOUSNESS?

WHAT IS THE SUBJECTIVE EXPERIENCE? NOBODY KNOWS



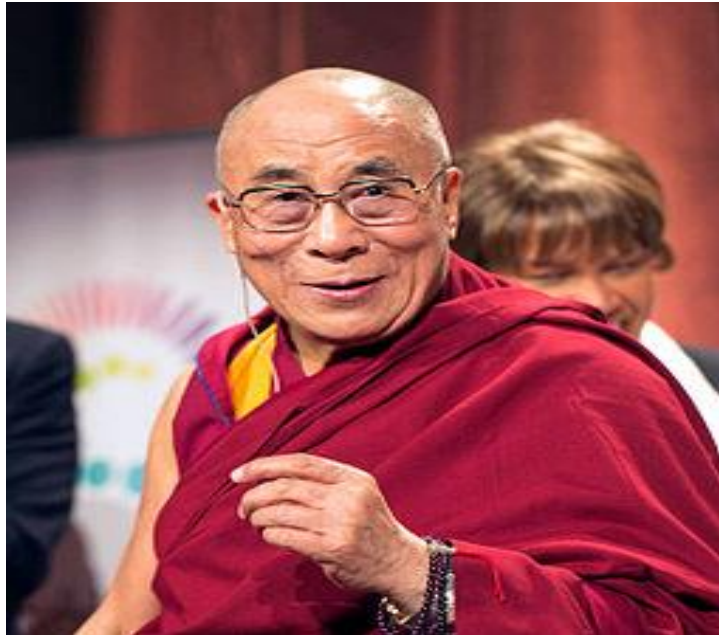
DISSOCIATION BETWEEN CORTICAL ACTIVITY AND COUSCIOUSNESS

- During episodes of unconsciousness, the child's brain is by no means inactive; it is actually hyperactive.
- The observation of similar firing rates between waking and slow-wave sleep represents both the most striking dissociation between cortical activity and consciousness.
- It indicates that the presence of “normal” firing levels in the cerebral cortex is not a sufficient criterion for consciousness.

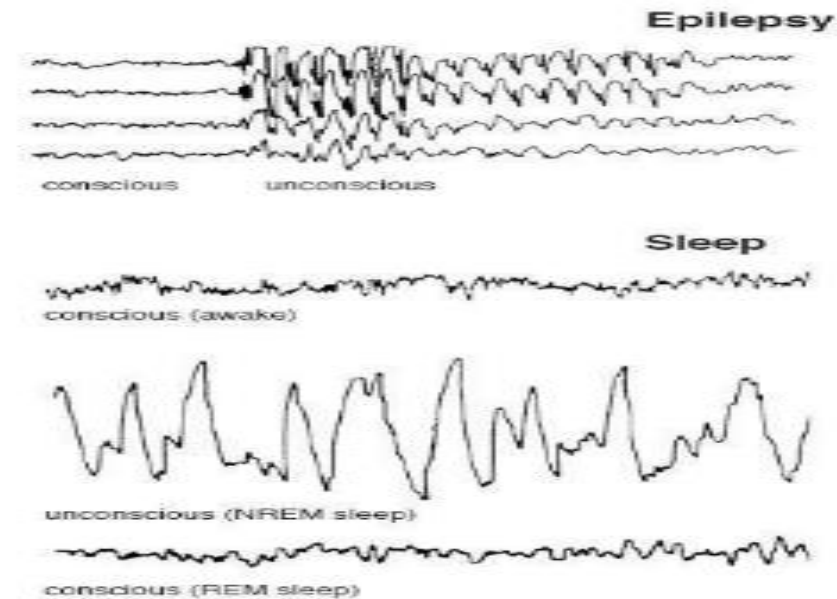


Edelman, Gerald. A Universe Of Consciousness: How
Matter Becomes Imagination

How Thoughts can change the Brain



Conscious Experience requires PATTERNS OF NEURAL ACTIVITY THAT ARE HIGHLY DIFFERENTIATED (Gerald Edelman)



Tenzin Gyatso (Dalai Lama): “What if the brain comes from consciousness instead of consciousness coming from the brain?”

HUMAN BRAIN VS CONSCIOUSNESS

An High Dimensional and Complex Neural Network



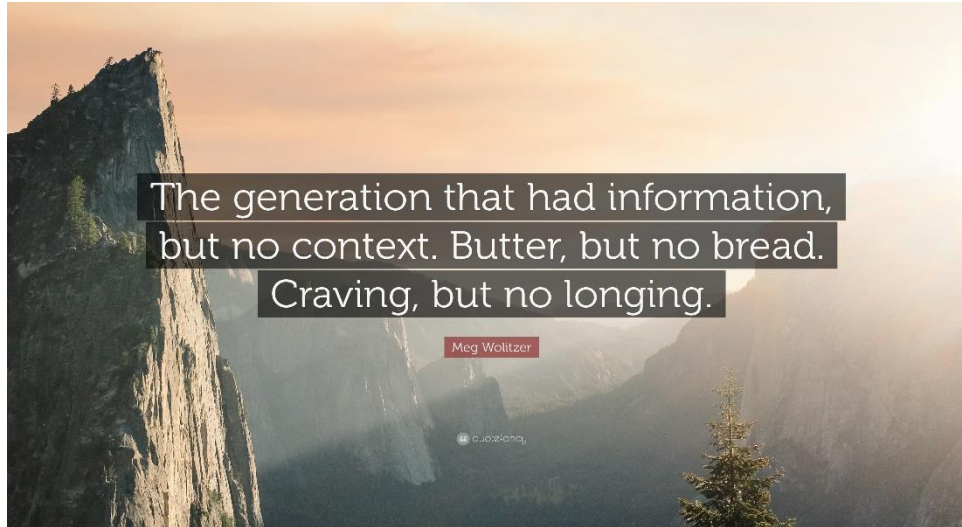
- The lack of a formal link between neural network structure and its emergent function has hampered our understanding of how the brain processes information.

Scientists find evidence of a Multidimensional Universe within the Brain



Reimann, M.W., Nolte, M., Scolamiero, M., Turner, K., Perin, R., Chindemi, G., .. & Markram, H. (2017). Cliques of neurons bound into cavities provide a missing link between structure and function. *Frontiers in computational neuroscience*, 11, 48.

MACHINE LEARNING VS CONSCIOUSNESS



Is this a Glass or a Pot?



Where is the Human-Like Thoughts Flow?

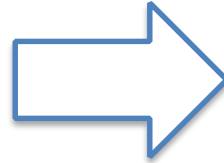
**Why and How Lea Seadol beats Alpha
Go Zero
once?**

**Even though all metrics say it is
impossible.**



MACHINE LEARNING LIMITS

Will ever Self Driving Car cope with Mumbai Traffic?

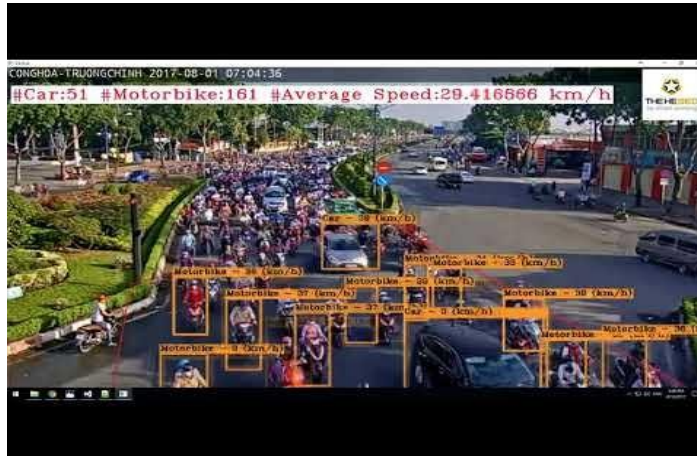


Will ever AI have context information ?

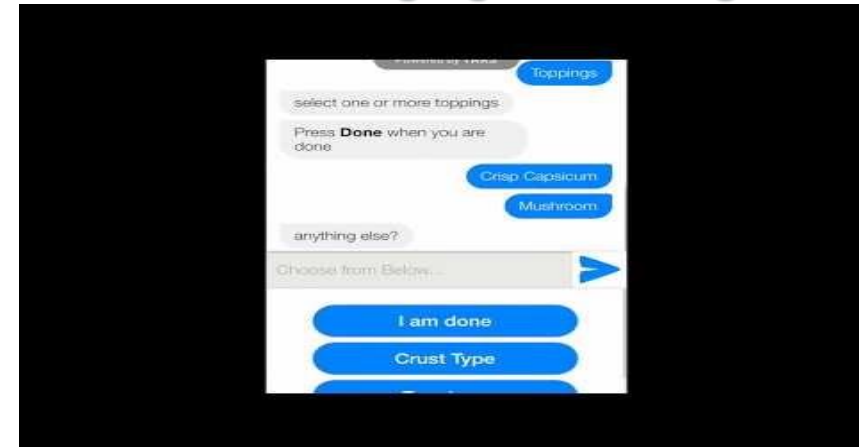


MACHINE LEARNING BREAKTHROUGHTS

Computer Vision



Natural Language Processing



A truly general Artificial Intelligence

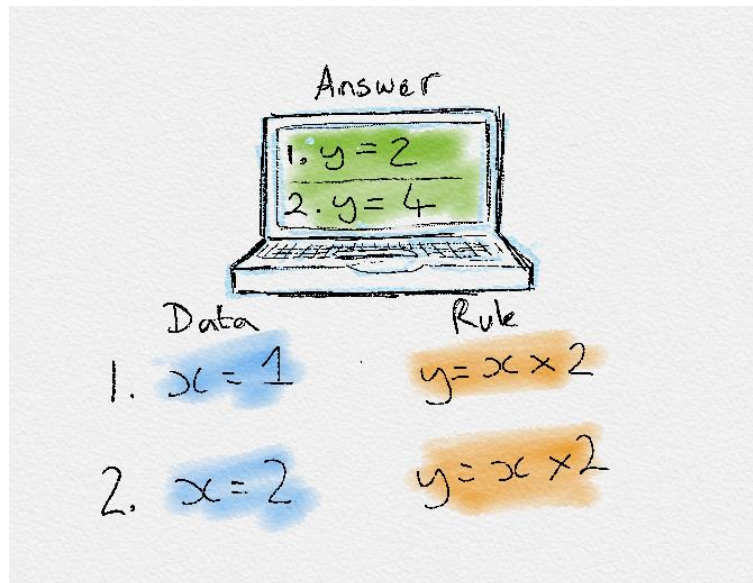


A Generative Artificial Intelligence

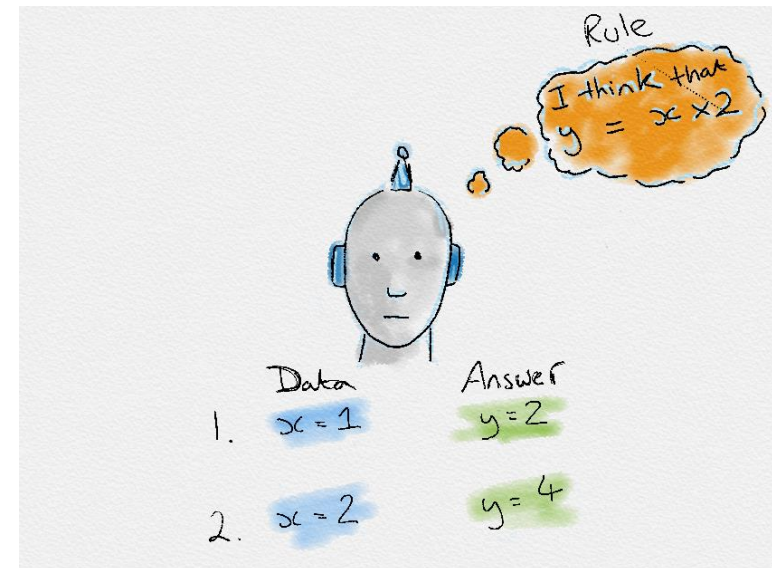


Traditional programming vs Machine Learning

- ✓ Machine learning is a subfield of computer science that is concerned with building algorithms which, to be useful, rely on a collection of examples of some phenomenon.
- ✓ Traditionally, software engineering combined human created rules with data to create answers to a problem. Instead, machine learning uses data and answers to discover the rules behind a problem.
- ✓ To learn the rules governing a phenomenon, machines have to go through a learning process, trying different rules and learning from how well they perform. Hence, why it's known as Machine Learning.

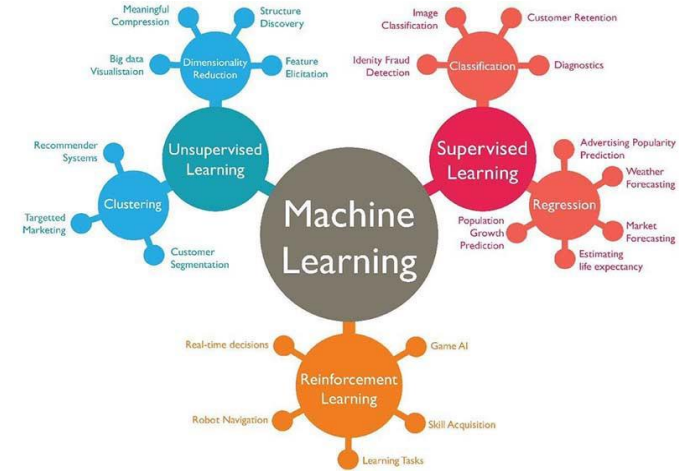


VS

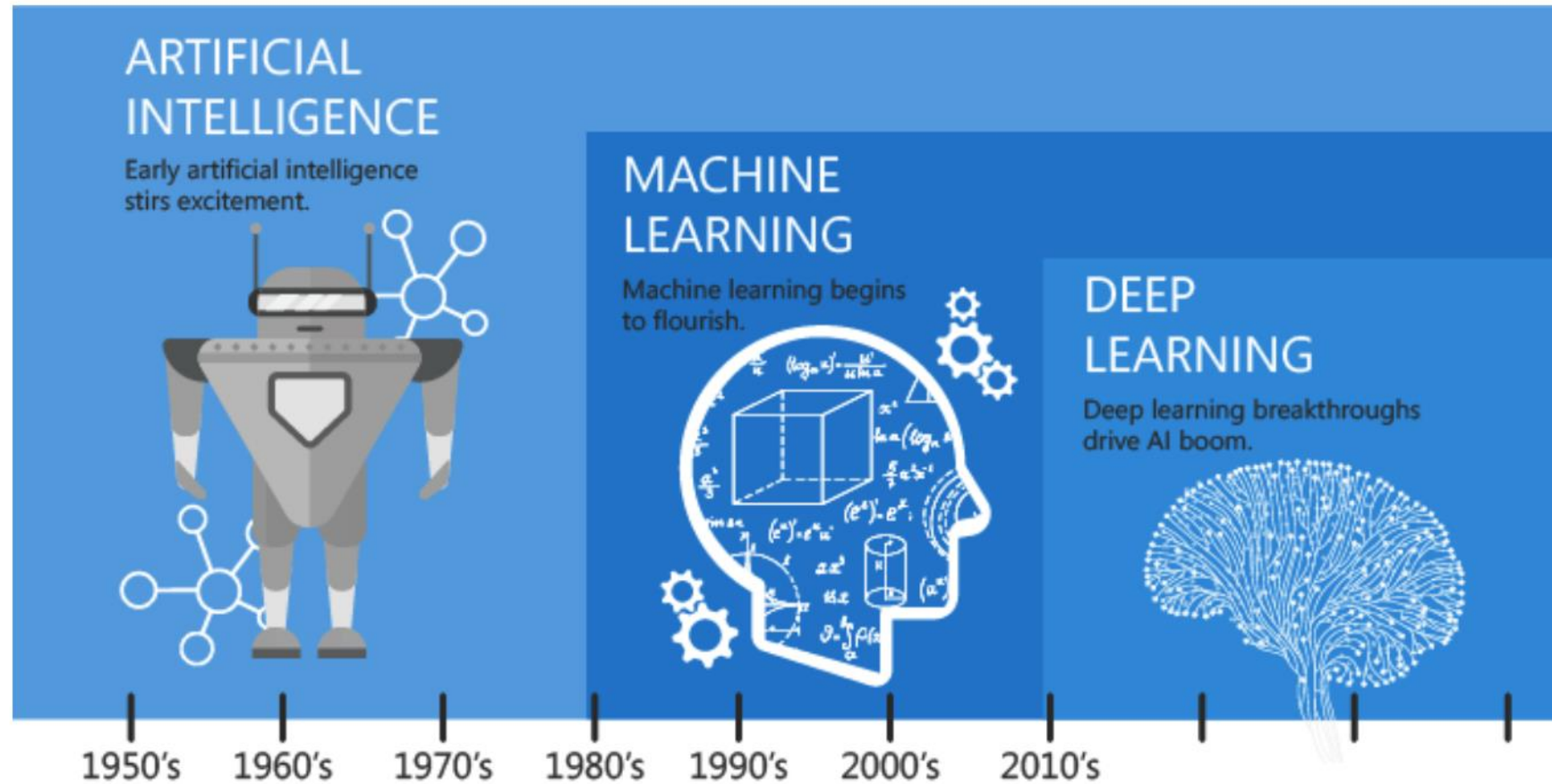


Types of Machine Learning

- ✓ There are broadly 4 types of Machine Learning namely,
 - ✓ Supervised learning
 - ✓ Unsupervised learning
 - ✓ Semi-supervised learning
 - ✓ Self-supervised learning
 - ✓ Reinforcement learning
- ✓ Each form of Machine Learning has differing approaches, but they all follow the same underlying process and theory.
- ✓ The **No Free Lunch theorem** is famous in Machine Learning. This theorem states that there is no single algorithm that works well for all types of tasks. Each task you attempt to solve has its own idiosyncrasies, meaning its own specificities and adversities to certain Machine Learning models. Therefore, there are many algorithms and approaches that adapt to each specificity and peculiarity of an individual problem.



A quick history of Machine Learning

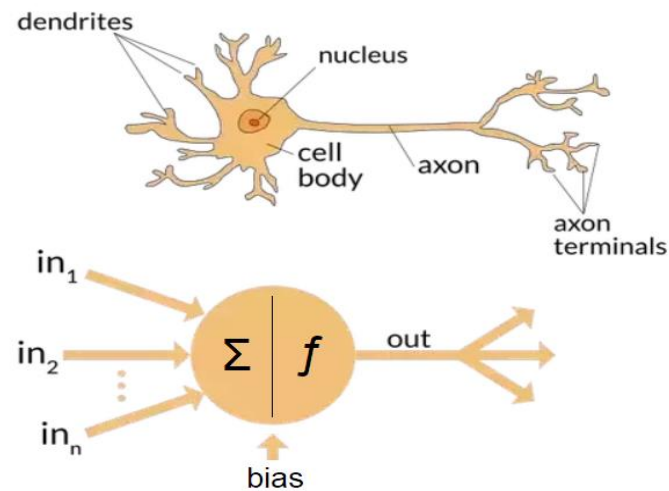


Since an early flush of optimism in the 1950's, smaller subsets of artificial intelligence - first machine learning, then deep learning, a subset of machine learning - have created ever larger disruptions.

ARTIFICIAL NEURAL NETWORKS

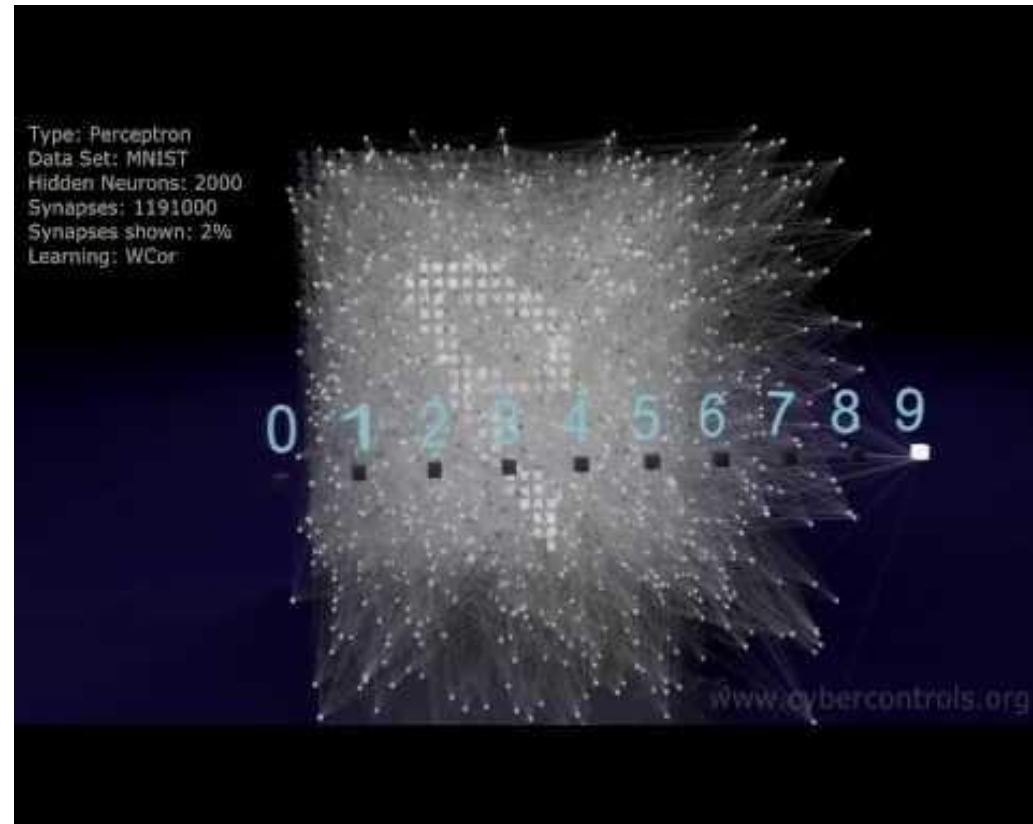
- **Artificial neural networks (ANN) or Connectionist Systems** are computing systems vaguely inspired by the biological neural networks that constitute animal brains (van Gerven & Bohte, 2018)

Biological Neuron Vs Artificial Neuron



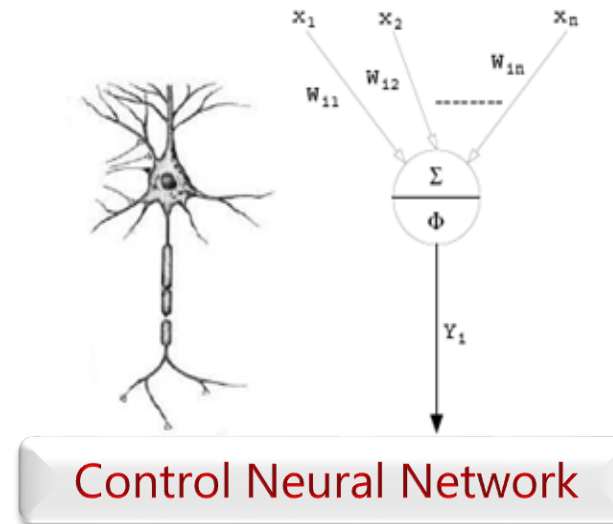
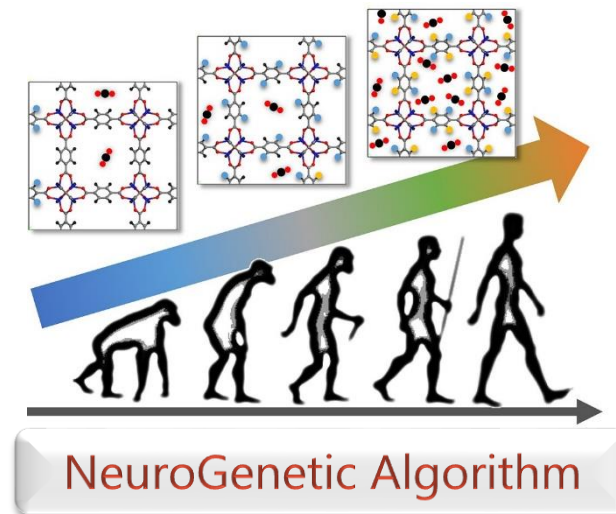
Neuron Activation

$$y_j = \Phi\left(\sum_{i=1}^N w_{ij}X_i - \theta_i\right)$$

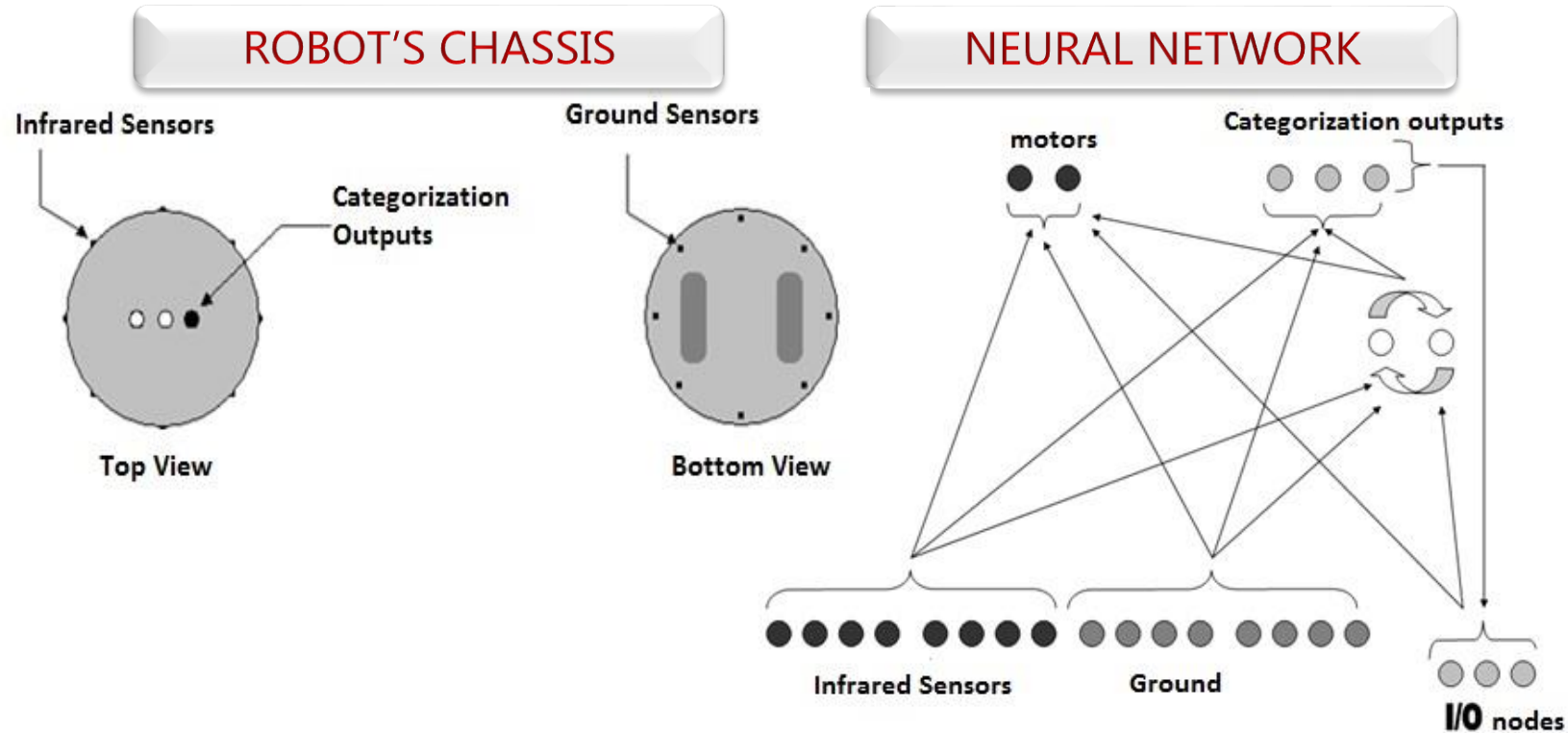


One traditional application ANN: Evolutionary robotics

IN ORDER TO OVERCOME THE PROBLEMS ASSOCIATED WITH THE ROBOTIC SYSTEM DECOMPOSITION OF TRADITIONAL APPROACHES (I.E. BEHAVIOR-BASED ROBOTICS), EVOLUTIONARY ROBOTICS CAN BE USED, WHERE THE ROBOTIC SYSTEM IS ABLE TO SELF-ORGANIZE
[NOLFI, S., FLOREANO, D ., 2002].



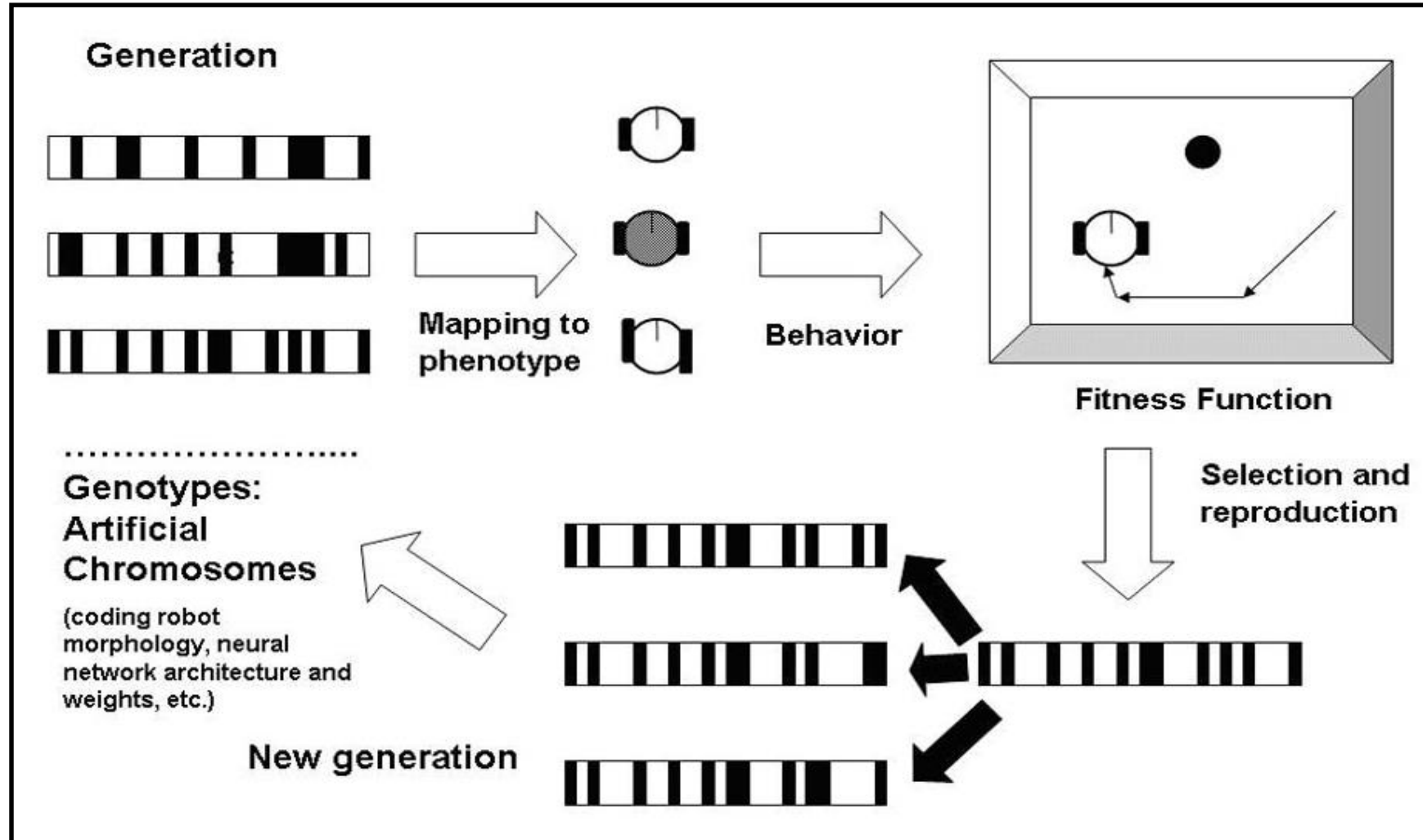
EXPERIMENTAL SETUP



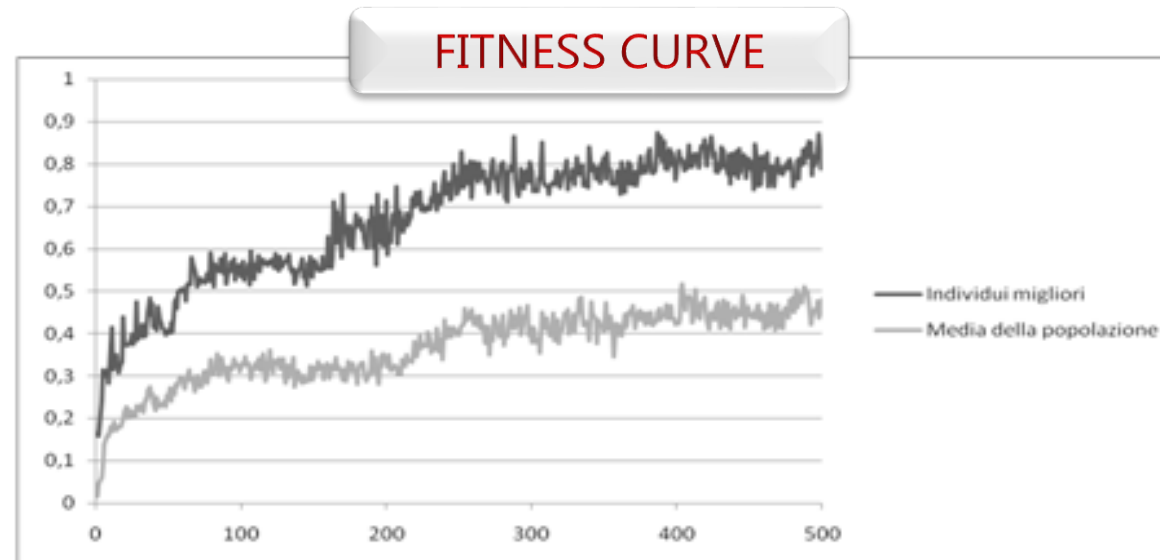
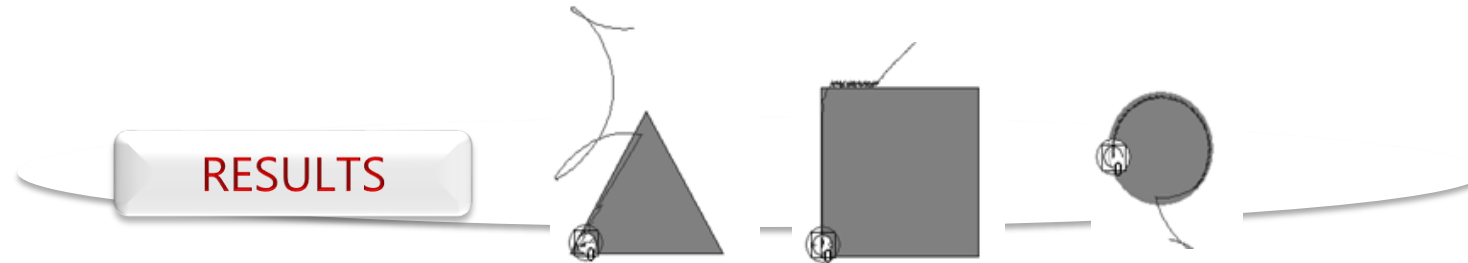
LEAKY Activation

$$A_j = t_j + \sum w_{ij} O_i, O_j = \delta_j O^{t-1} + (1 - \delta_j) \left(1 + \frac{1}{e^{A_j}} \right), 0 \leq \delta_j \leq 1$$

Evolutionary robotics



EXPERIMENTAL SETUP N.1

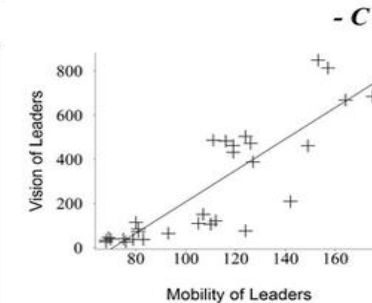
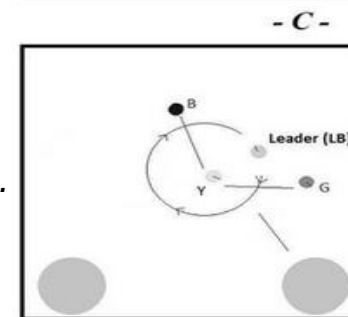
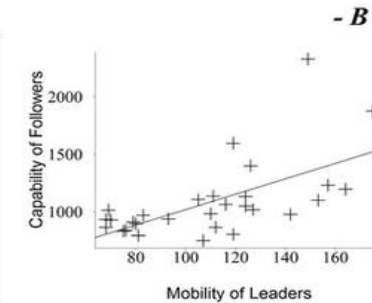
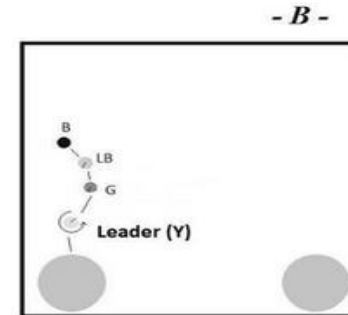
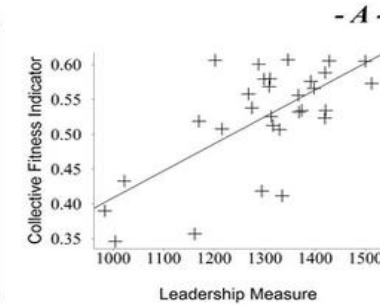
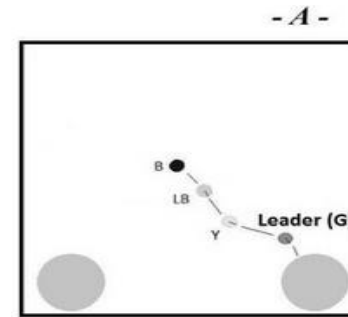


Emergence of Leadership in Robots

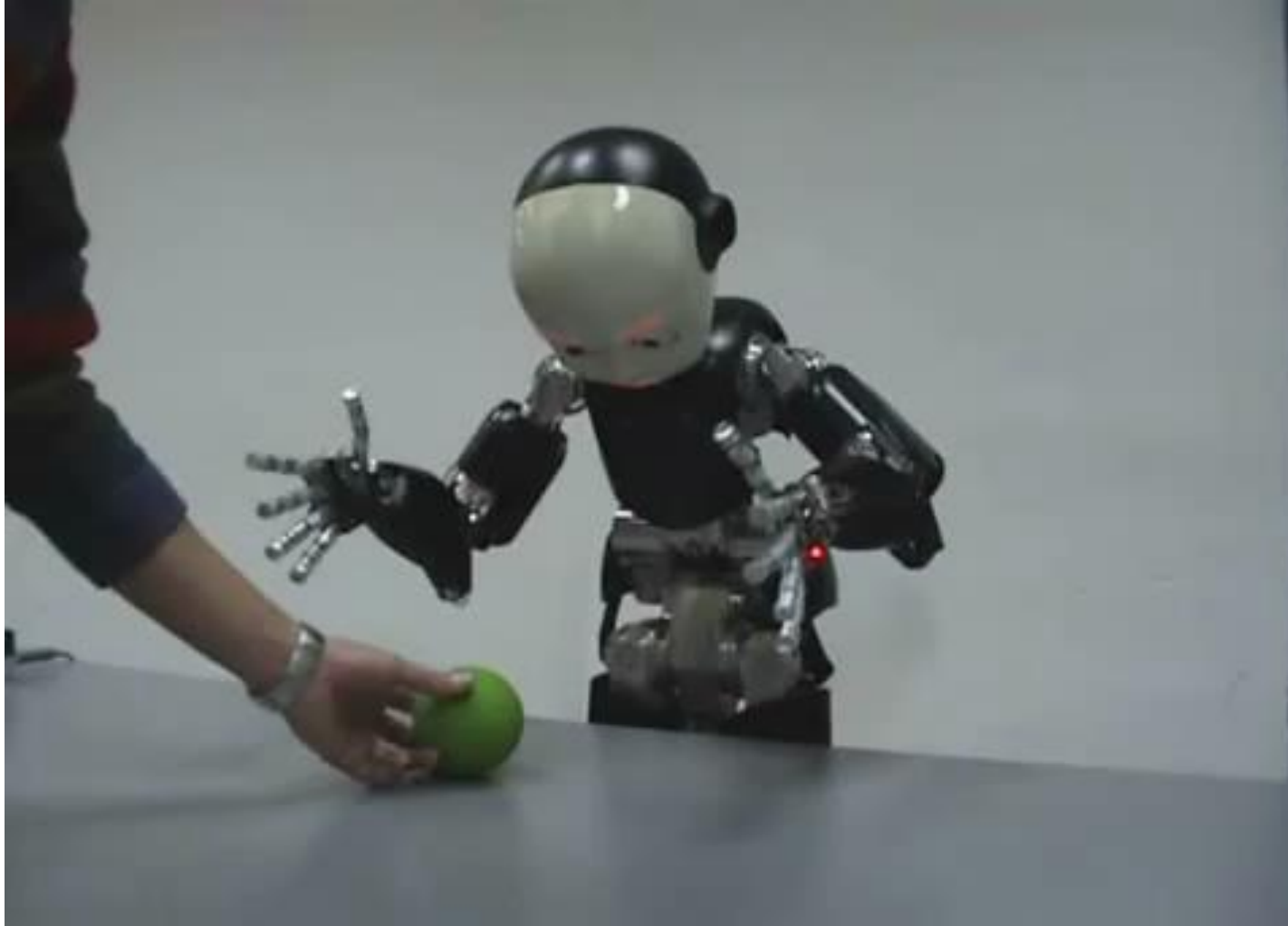
- Behavioural and quantitative analysis indicate that a form of leadership emerges
- Groups with a leader are more effective than groups without.
- The most skilled individuals in a group tend to be the leaders.
- Further analysis reveals the emergence of different “styles” of leadership (active and passive).

- A - *Passive Leadership.* - B - *Weak Active Leadership.*
- C - *Strong Active Leadership.*

- (Pugliese, et. Al., 2015)



MAY Robotics help to understand social and psychological problems?



DEEP LEARNING: Neural Networks become more effective

In recent years **Deep Neural Networks** have achieved noticeably breakthroughs in research (Bengio, 2009). This new methodology dealing with deep neural networks and their training algorithms is called “*Deep Learning*”. So far, in all the experiments, the resulting performances were many magnitudes better than other machine learning techniques available.



GOOGLE DATACENTER

1,000 CPU Servers
2,000 CPUs • 16,000 cores

600 kWatts
\$5,000,000

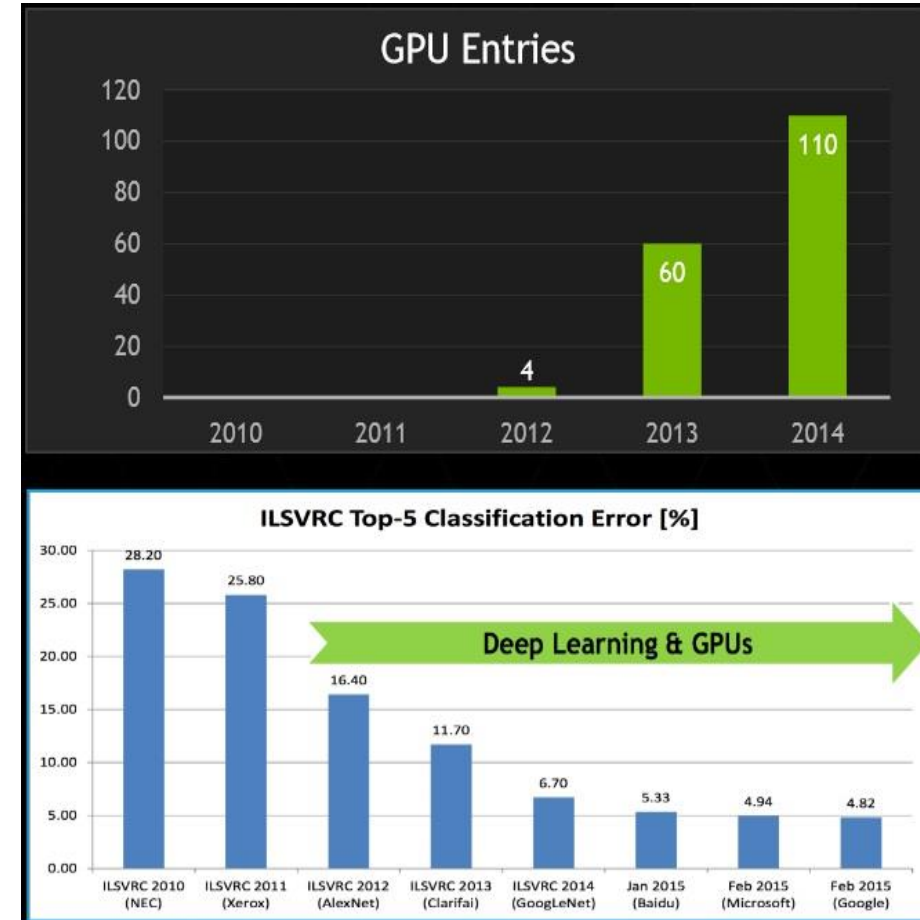
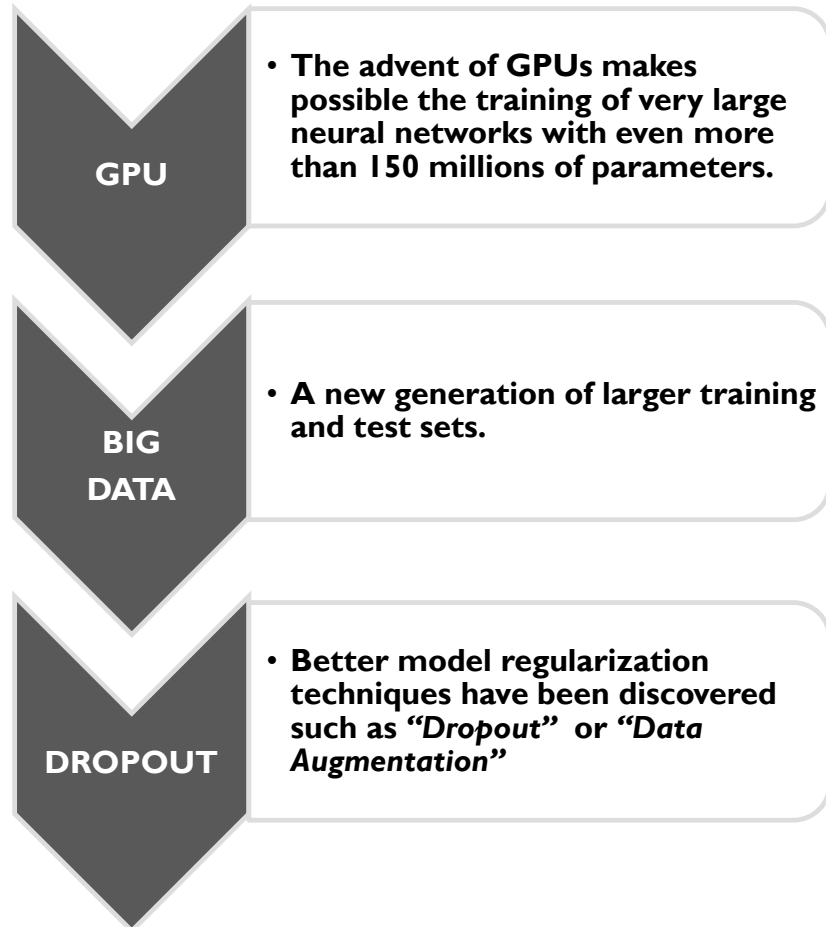


STANFORD AI LAB

3 GPU-Accelerated Servers
12 GPUs • 18,432 cores

4 kWatts
\$33,000

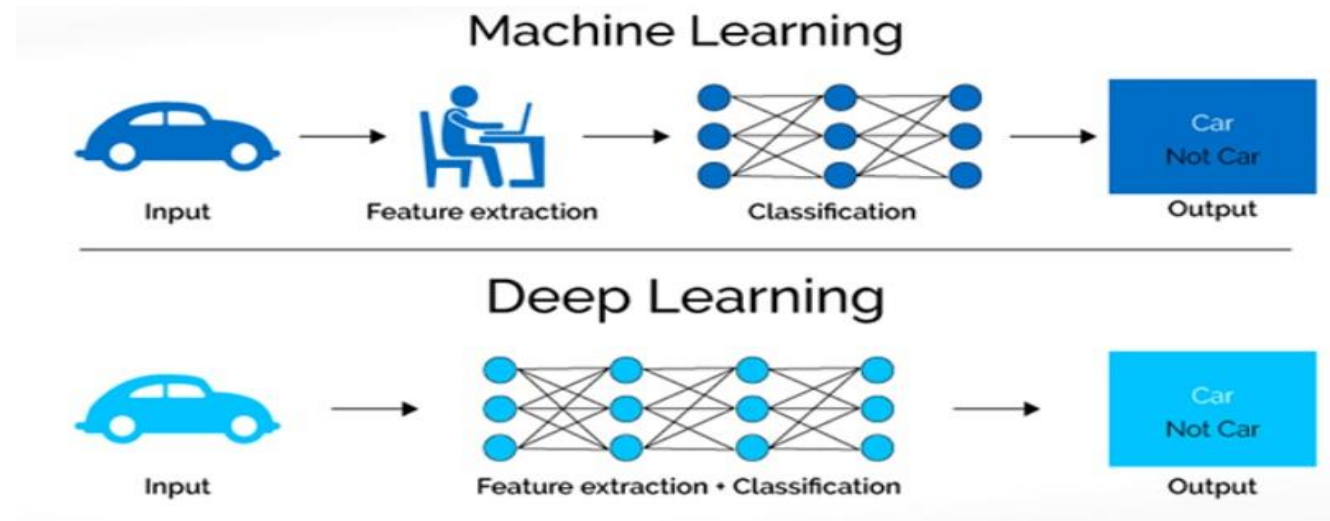
DEEP LEARNING: a cutting-edge approach to Computer Vision and NLP



Deep Learning is Machine Learning

Deep Learning refers to algorithms that automatically 'model' high-level abstractions in data

- i. here 'model' means: define, find, recognize and exploit
- ii. here 'automatically' means: directly from data, without hinging upon handcrafted, task-specific features.



Why Deep Learning over-performed traditional statistics models?



- “Deep Learning” approaches can be **end-to-end trained** without a task-specific feature engineering.
- **These models are scalable:** adding GPUs they can be trained faster.
- *“Deep Learning is killing every problem in AI”* (Elizabeth Gibney, 2016)
- Basically, **statistics is not able to deal with very high dimensionalities** of data as Deep Learning does.

28

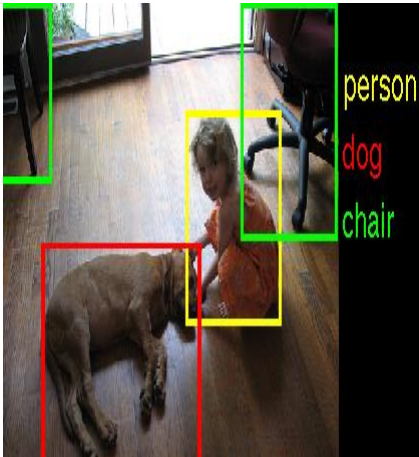
Computer Vision: Where does Traditional Statistics fail?



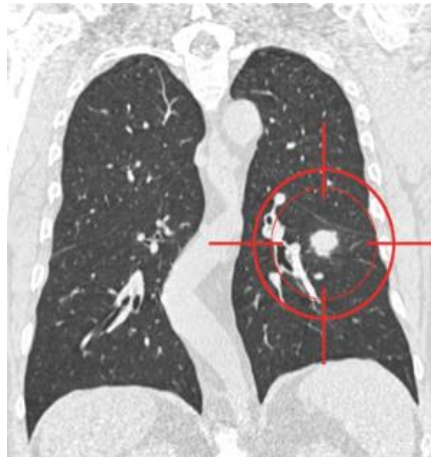
- **Computer Vision** is an interdisciplinary field that deals with the way algorithms can be made for gaining high-level understanding from digital images or videos.
- **Statistical methods** are not always welcome in computer vision.
- Statistical methods seem **not scaling up** to the challenges of computer vision problems (Chellappa, R., 2012).

Why does Computer Vision matter so much?

EVERYDAY LIFE



BIOMEDICAL IMAGES



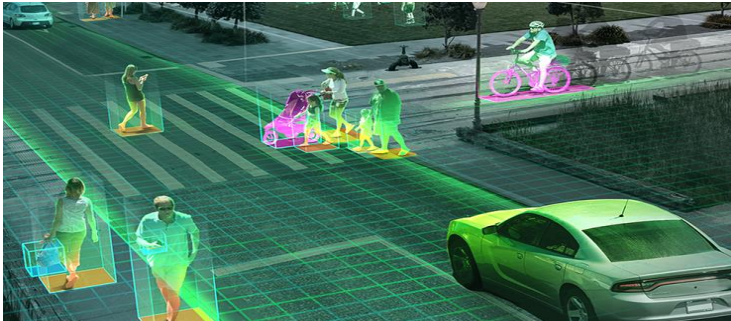
- A new study proves the **relationship** between **Vision capabilities and Intelligence** (Tsukahara et al., 2016).
- Seemingly, **human beings** have one of **the most complex vision systems** within the **animal realm** and this fact would have fostered **the evolution of human intelligence**.
- In other words, **Computer Vision** needs **human-like intellectual capabilities** in order to achieve the vision performances required by humans for every-day applications.

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Why does Computer Vision matter so much?



- **A new generation of machines** might accomplish typical human tasks such as recognizing and moving objects, driving cars, cultivating fields, cleaning streets, city garbage collecting, etc.



Why does Computer Vision matter so much?

Computer Vision for Recycling



- **Smart Garbage Bins** could classify waste and differentiate it automatically. This might reduce the enormous impact of human errors.
- As far as food **sustainability** is concerned, a first version of a classifier might enable a **restaurant** or a shop to recognize whether there is still a return of residual food from the production process. Afterwards, by exploiting other sub-classifiers we could catch the opportunity of regenerating this residual food saving resources.
- **AI applications** which can optimize all the **flow of interactions** with a **restaurant**, decreasing the process costs and increasing sustainability (*«Deep Learning al servizio del riciclo»*, *Intervista su Wired, Pugliese, F.*). Thanks to **Artificial Intelligence**, in one work, some researchers have generated menus of some restaurants analyzing all **non-structured data** from customers online comments.

Why does Computer Vision matter so much?

Image Classification Problem

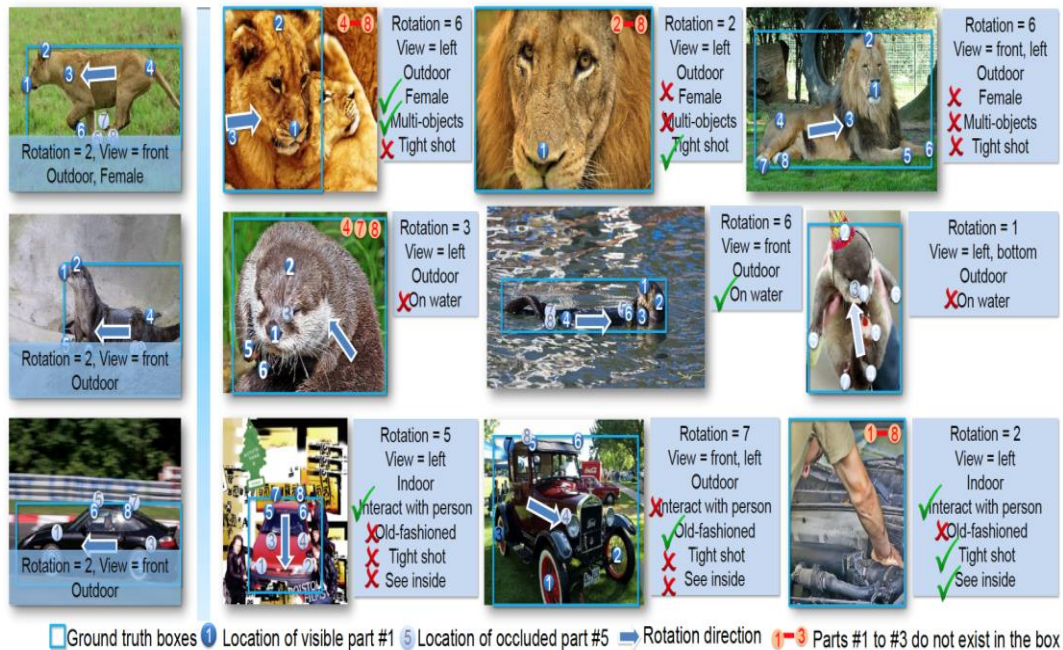


What we see

What computers see

```
08 02 22 97 38 15 00 40 00 75 04 05 07 78 52 12 50 77 91 08
49 49 99 40 17 81 18 57 60 87 17 40 98 43 69 48 04 56 62 00
81 49 31 73 55 79 14 29 93 71 40 67 53 88 30 03 49 13 36 65
52 70 95 23 04 60 11 42 69 24 68 56 01 32 56 71 37 02 36 91
22 31 16 71 51 67 63 89 41 92 36 54 22 40 40 28 66 33 13 80
24 47 32 60 99 03 45 02 44 75 33 53 78 36 84 20 35 17 12 50
32 98 81 28 64 23 67 10 26 38 40 67 59 54 70 66 18 38 64 70
67 26 20 68 02 62 12 20 95 63 94 39 63 08 40 91 66 49 94 21
24 55 58 05 66 73 99 26 97 17 78 78 96 83 14 88 34 89 63 72
21 36 23 09 75 00 76 44 20 45 35 14 00 61 33 97 34 31 33 95
78 17 53 28 22 75 31 67 13 94 03 80 04 62 16 14 09 53 56 92
16 39 05 42 96 35 31 47 55 58 88 24 00 17 54 24 36 29 85 57
86 56 00 48 35 71 89 07 05 44 44 37 44 60 21 58 51 54 17 58
19 80 81 68 05 94 47 49 28 73 92 13 86 32 17 77 04 89 55 40
04 52 08 83 97 35 99 16 07 97 57 32 16 26 26 79 33 27 98 66
88 36 68 87 57 62 20 72 03 46 33 67 46 55 12 32 63 93 53 69
04 42 16 73 38 25 39 11 24 94 72 18 08 46 29 32 40 62 76 36
20 69 36 41 72 30 23 88 34 62 99 69 82 67 59 85 74 04 36 16
20 73 35 29 78 31 90 01 74 31 49 71 48 86 81 16 23 57 05 54
01 70 54 71 83 51 54 69 16 92 33 48 61 43 52 01 89 19 67 48
```


Computer Vision Datasets and Competitions



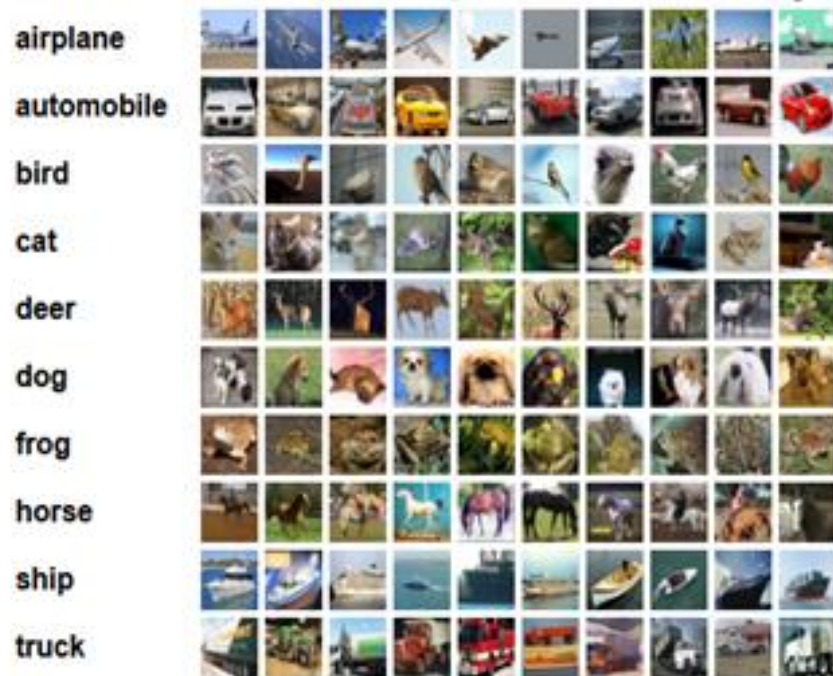
- **ImageNet:** ImageNet is a dataset of over **15** million labeled high-resolution images belonging to roughly **22,000** categories.
- Since **2010** a competition called «ImageNet Large-Scale Visual Recognition Challenge (ILSVRC)» uses a subset of ImageNet with roughly **1000** images in each of **1000** categories.

Train Set: 1.2 million

Validation Set: 50,000

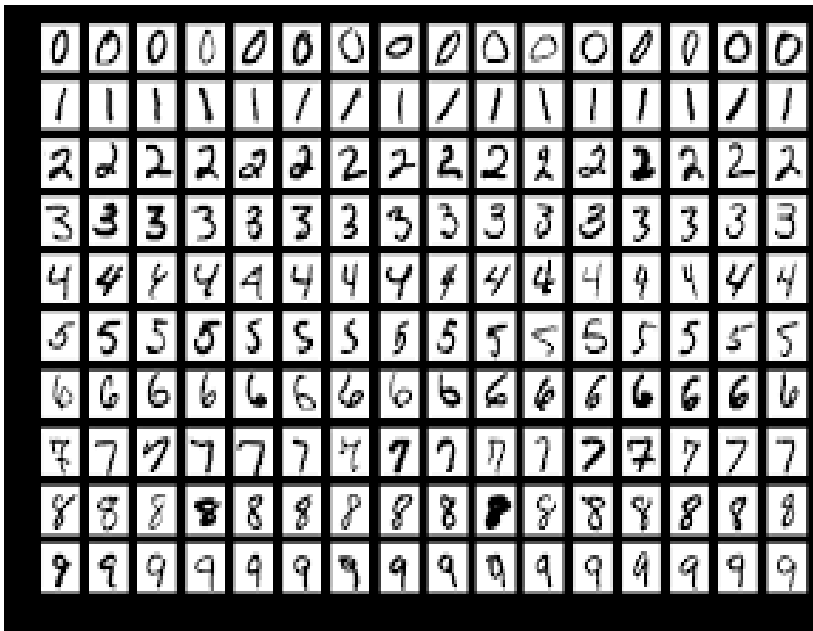
Test set: 150,000

Computer Vision Datasets and Competitions



- The **CIFAR-10** dataset consists of **60,000 32x32** coloured images divided into **10 classes**, with **6000 images per class**. There are **50,000 training** images and **10,000 test** images.
- The dataset is divided into **five training batches** and one test batch, each with **10,000** images. The test batch contains exactly 1000 randomly-selected images from each class. The training batches contain the remaining images in random order, but some training batches may contain more images from one class than another. Between them, the training batches contain exactly 5000 images from each class.
- The classes are completely **mutually exclusive**. There is **no overlap** between **automobiles** and **trucks**. "Automobile" includes sedans, SUVs, things of that sort. "Truck" includes only big trucks. Neither includes pickup trucks.

Computer Vision Datasets and Competitions



- The **MNIST** database of handwritten digits has a training set of **60,000** examples, and a test set of **10,000** examples.
- It is a subset of a larger set available from **NIST**. The digits have been size-normalized and centered in a fixed-size image.
- It is a good database for people who want to try learning techniques and pattern recognition methods on real-world data while spending minimal efforts on **preprocessing** and **formatting**.
- The images contain **grey levels** as a result of the **anti-aliasing technique** used by the normalization algorithm. The images were centered in a **28x28** image by computing the center of mass of the pixels, and translating the image so as to position this point at the center of the **28x28** field.

Computer Vision Datasets and Competitions



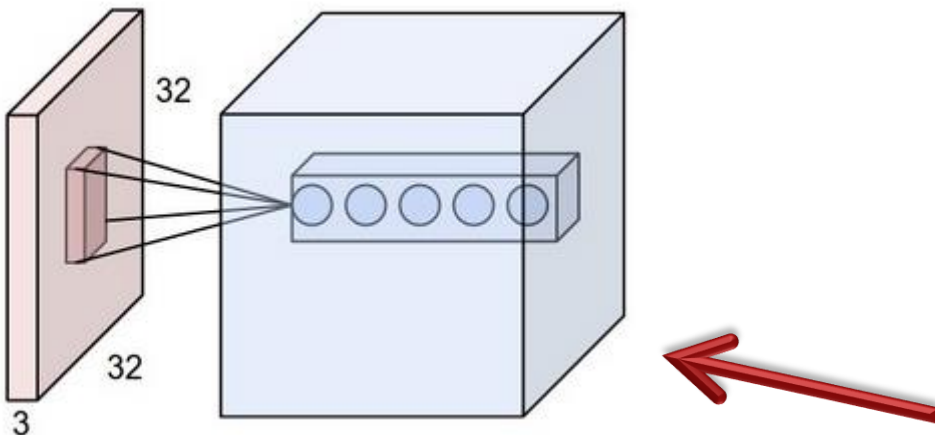
- **Fashion MNIST** is a dataset of **Zalando's** article images consisting of a training set of **60,000** images and a test set of **10,000** examples.
- Each example is a **28 x 28** grayscale image associated with a label from **10 classes**.
- Reasons for replacing MNIST according to **Zalando** opinions:
 - **MNIST is too easy.** Convolutional nets can achieve 99.7% on **MNIST**. Classic machine learning algorithms can also achieve 97% easily.
 - **MNIST is overused and cannot represent modern CV tasks**, as noted by deep learning expert/Keras author **François Chollet**.

Convolutional Neural Networks (ConvNets or CNNs)

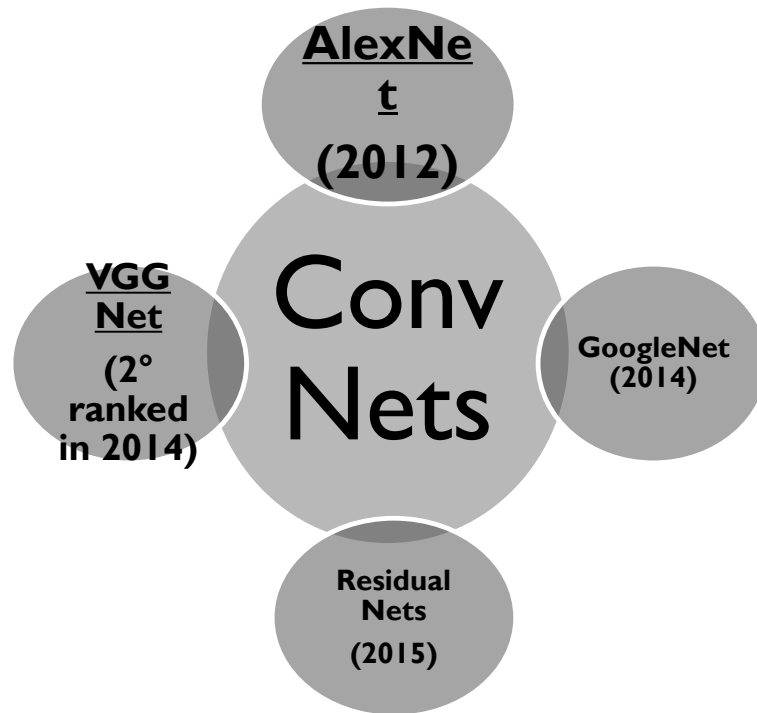
- **Convolutional Neural Networks (CNN)** are biologically-inspired variants of **MLPs**. We know the visual cortex contains a complex arrangement of cells (**Hubel, D. and Wiesel, T., 1968**). These cells are sensitive to small sub-regions of the visual field, called a *receptive field*. Other layers are: **RELU layer**, **Pool Layer**. Typical CNNs settings are: a) **Number of Kernels (Filters)**, b) **Receptive Field size**, b) **Padding**, c) **Stride**. These parameters are tied by the following equation:

$$(W - F + 2P) / S + 1$$

- Each neuron in the convolutional layer is connected only to a local region in the input volume spatially. In this case there are 5 neurons along the depth all looking at the same region.



Former Successful Deep Neural Networks



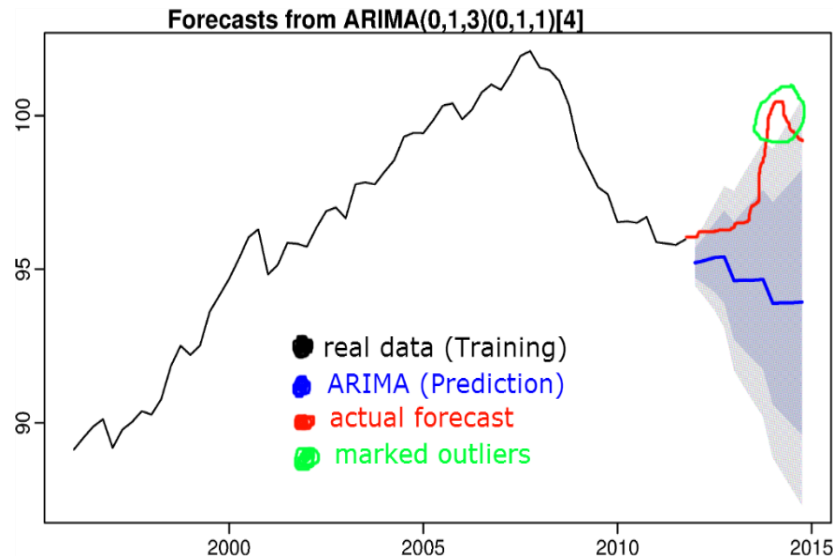
Traditional issues with Convolutional Layers:

- Wide convolutional layers determine overfitting and vanishing gradient problem with the Solver (SGD, Adam, etc).
- Low depth architectures produce raw features (need to push the depth forward).

Model Regularization :

- Dropout - Co-adaptation and Models Ensemble (Srivastava, N. et al., 2014).
- Weight penalty L1/L2
- Data Augmentation (crop, flip, rotation, ZCA whitening, etc.)

Deep Learning for Time - Series Prediction



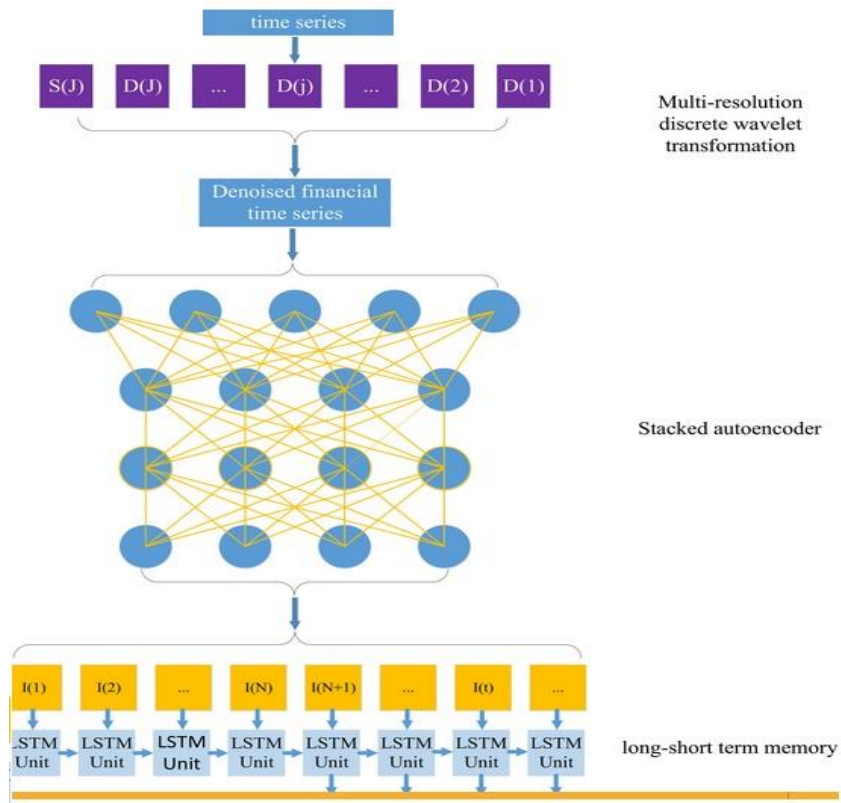
- The application of **Deep Learning** approaches to time-series prediction has received a great deal of attention from both entrepreneurs and researchers. Results show that deep learning models outperform other statistical models in predictive accuracy (**Bao, et al., 2017**).

$$X_t - \alpha_1 X_{t-1} - \dots - \alpha_{p'} X_{t-p'} = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q},$$

$$\left(1 - \sum_{i=1}^{p'} \alpha_i L^i\right) X_t = \left(1 + \sum_{i=1}^q \theta_i L^i\right) \varepsilon_t$$

- The application of classic time series models, such as **Auto Regressive Integrated Moving Average (ARIMA)**, usually requires strict assumptions regarding the distributions and stationarity of time series. For complex, non-stationary and noisy time-series it is necessary for one to know the properties of the time series before the application of classic time series models (**Bodyanskiy and Popov, 2006**). Otherwise, the forecasting effort would be ineffective.

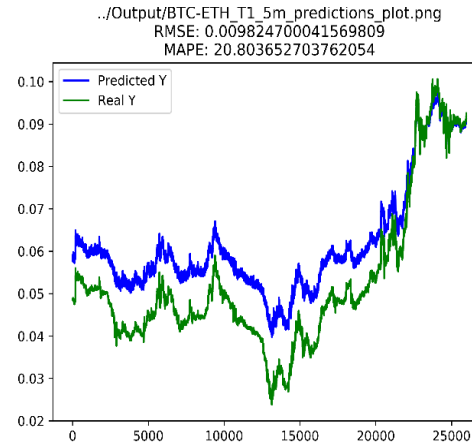
Advantages of Artificial Neural Networks (ANNs) in Time-Series Prediction



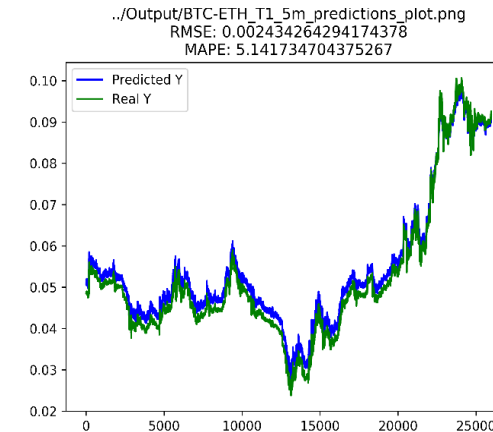
- However, by using ANNs, a priori analysis as ANNs do not require prior knowledge of the time series structure because of their black-box properties (**Nourani, et al., 2009**).
- Also, the impact of the stationarity of time series on the prediction power of ANNs is quite small. It is feasible to relax the stationarity condition to non-stationary time series when applying ANNs to predictions (**Kim, et al., 2004**).
- ANNs allow **multivariate time-series forecasting** whereas classical linear methods can be difficult to adapt to multivariate or multiple input forecasting problems.

Bitcoin BTC-ETH exchange Time Series Prediction – 5 mins (Poloniex)

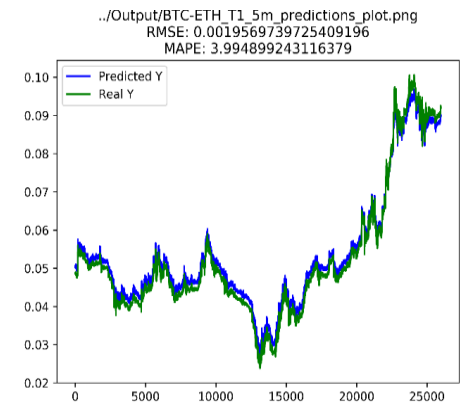
Test Set : 10%



1 Epoch



100 Epochs



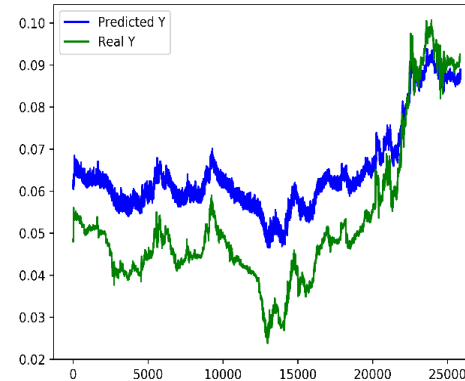
1000 Epochs

Bitcoin BTC-ETH exchange Time Series Prediction – 4 hours (Poloniex)

Test Set : 10%

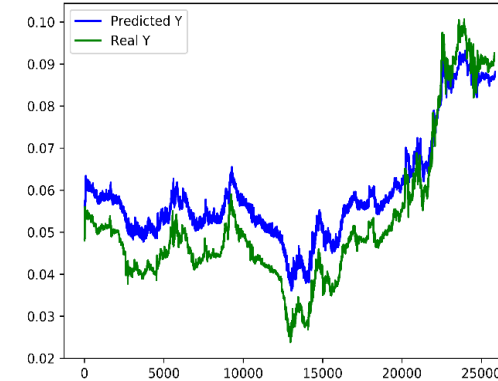


../Output/BTC-ETH_T1_04h_full_predictions_plot.png
RMSE: 0.013678025667312696
MAPE: 29.04875874519348



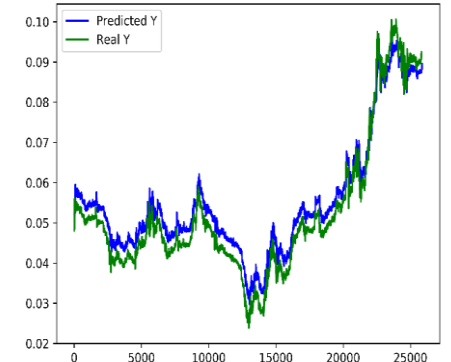
1 Epoch

../Output/BTC-ETH_T1_04h_full_predictions_plot.png
RMSE: 0.008203839284446948
MAPE: 17.33316034078598



100 Epochs

../Output/BTC-ETH_T1_04h_full_predictions_plot.png
RMSE: 0.0039319158198738895
MAPE: 7.922710478305817



1000 Epochs

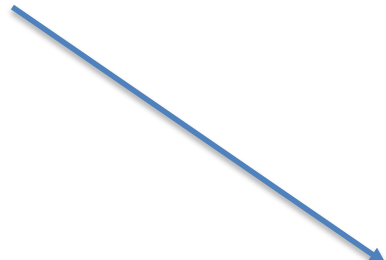
Data Preparation : From Time-Series a Supervised Problem

example

i	features					label
	$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	...	$x^{(j)}$	y
1	75	1.77	25	...	...	yes
2	68	1.65	32	...	...	X
3	57	1.60	27	...	...	no
4	84	1.90	41	...	...	X
...	...	...	...	...	...	...
N	...	...	...	...	...	...

OKPEDIA

1	0
2	1
3	2
4	3
5	4
6	5
7	6
8	7
9	8
10	9

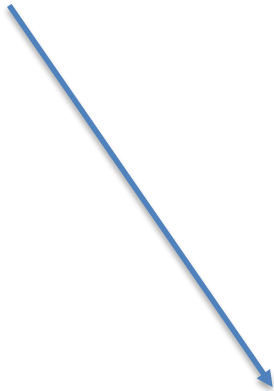


1	X, y
2	1 2
3	2, 3
4	3, 4
5	4, 5
6	5, 6
7	6, 7
8	7, 8
9	8, 9

Data Preparation : Pandas shift function



```
1 from pandas import DataFrame
2 df = DataFrame()
3 df['t'] = [x for x in range(10)]
4 df['t-1'] = df['t'].shift(1)
5 print(df)
```



	t	t-1
1	0	NaN
2	1	0.0
3	2	1.0
4	3	2.0
5	4	3.0
6	5	4.0
7	6	5.0
8	7	6.0
9	8	7.0
10	9	8.0

Data Preparation : Multi-Step Univariate Analysis

```
1 data = series_to_supervised(values, 3)
```



1		var1(t-3)	var1(t-2)	var1(t-1)	var1(t)
2	3	0.0	1.0	2.0	3
3	4	1.0	2.0	3.0	4
4	5	2.0	3.0	4.0	5
5	6	3.0	4.0	5.0	6
6	7	4.0	5.0	6.0	7
7	8	5.0	6.0	7.0	8
8	9	6.0	7.0	8.0	9

Data Preparation : Multi-Step or Sequence Forecasting

```
1 data = series_to_supervised(values, 2, 2)
```



1		var1(t-2)	var1(t-1)	var1(t)	var1(t+1)
2	2	0.0	1.0	2	3.0
3	3	1.0	2.0	3	4.0
4	4	2.0	3.0	4	5.0
5	5	3.0	4.0	5	6.0
6	6	4.0	5.0	6	7.0
7	7	5.0	6.0	7	8.0
8	8	6.0	7.0	8	9.0

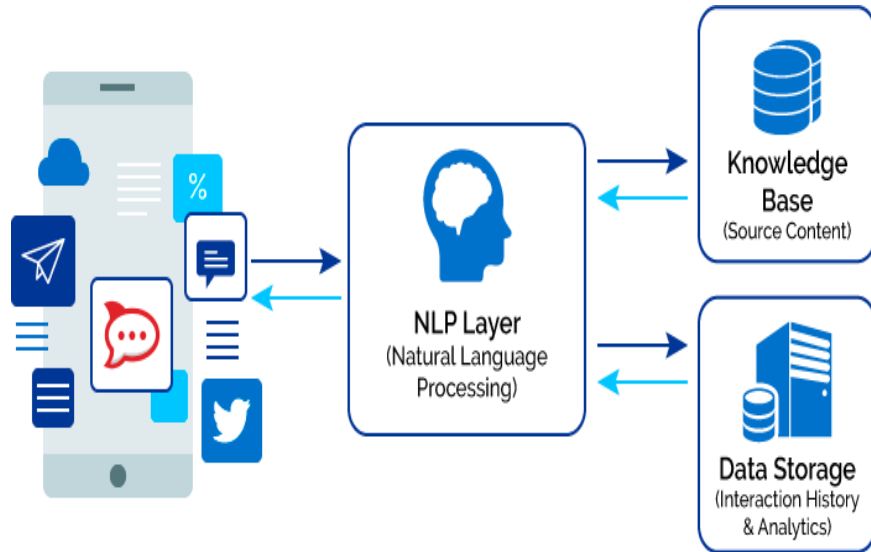
Data Preparation : Multivariate Forecasting



1		var1(t-1)	var2(t-1)	var1(t)	var2(t)
2	1	0.0	50.0	1	51
3	2	1.0	51.0	2	52
4	3	2.0	52.0	3	53
5	4	3.0	53.0	4	54
6	5	4.0	54.0	5	55
7	6	5.0	55.0	6	56
8	7	6.0	56.0	7	57
9	8	7.0	57.0	8	58
10	9	8.0	58.0	9	59

1		var1(t-1)	var2(t-1)	var1(t)	var2(t)	var1(t+1)	var2(t+1)
2	1	0.0	50.0	1	51	2.0	52.0
3	2	1.0	51.0	2	52	3.0	53.0
4	3	2.0	52.0	3	53	4.0	54.0
5	4	3.0	53.0	4	54	5.0	55.0
6	5	4.0	54.0	5	55	6.0	56.0
7	6	5.0	55.0	6	56	7.0	57.0
8	7	6.0	56.0	7	57	8.0	58.0
9	8	7.0	57.0	8	58	9.0	59.0

Textual Big Data alias The problem of the Natural Language Processing - NLP



- Understanding **complex language utterances** is one of the **hardest challenge** for Artificial Intelligence (AI) and Machine Learning (ML).
- **NLP** is everywhere because people communicate most everything: web search, advertisement, emails, customer service, etc.

Deep Learning and NLP



- “Deep Learning” approaches have obtained very high performance across many different **NLP** tasks. These models can often be trained with a single **end-to-end model** and do not require traditional, task-specific feature engineering.

(Stanford University School Of Engineering – CS224D)

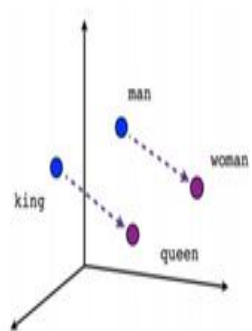
- **Natural language Processing** is shifting from statistical methods to **Neural Networks**.

7 NLP applications where Deep Learning achieved «state-of-art» performance

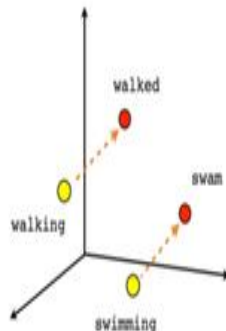


- **1 Text Classification:** Classifying the topic or theme of a document (i.e. Sentiment Analysis).
- **2 Language Modeling:** Predict the next word given the previous words. It is fundamental for other tasks.
- **3 Speech Recognition:** Mapping an acoustic signal containing a spoken natural language utterance into the corresponding sequence of words intended by the speaker.
- **4 Caption Generation:** Given a digital image, such as a photo, generate a textual description of the contents of the image.
- **5 Machine Translation:** Automatic translation of text or speech from one language to another, is one [of] the most important applications of NLP.
- **6 Document Summarization:** It is the task where a short description of a text document is created.
- **7 Question Answering:** It is the task where the system tries to answer a user query that is formulated in the form of a question by returning the appropriate noun phrase such as a location, a person, or a date. (i.e. Who killed President Kennedy? Oswald)

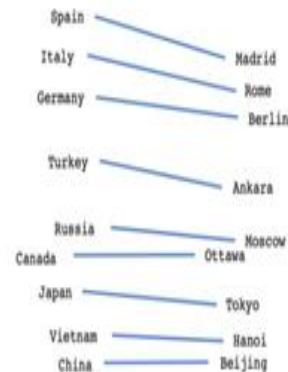
Word Embedding & Language Modeling



Male-Female



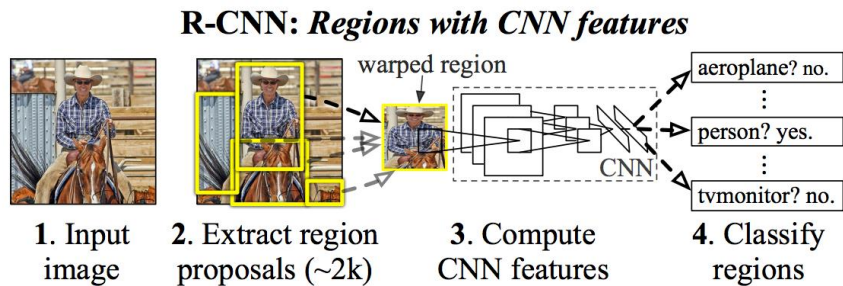
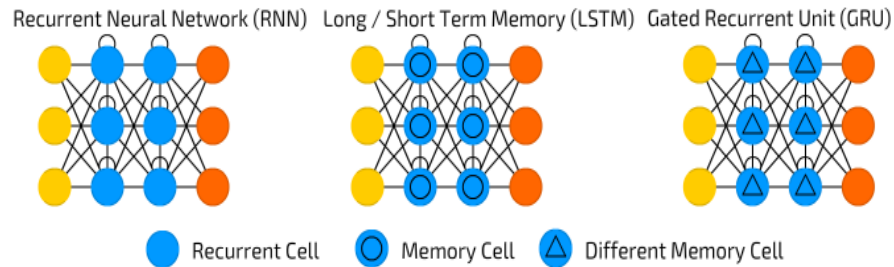
Verb tense



Country-Capital

- **Word embedding** is the collective name for a set of language modeling and feature learning techniques for natural language processing (**NLP**) where words or sentences from the vocabulary are mapped to vectors of real numbers.
- These vectors are semantically correlated by metrics like **cosine distance**.

Text Classification Models

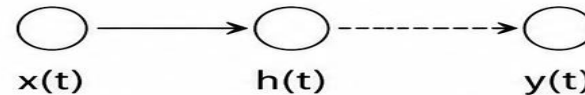


- RNN, LSTM, GRU, ConvLstm, RecursiveNN, RNTN, RCNN
- The modus operandi for text classification involves the use of a pre-trained **word embedding** for **representing words** and a **deep neural networks** for learning how to **discriminate documents** on classification problems.
- The **non-linearity of the NN** leads to superior classification accuracy.

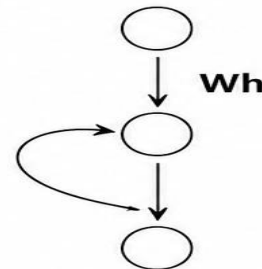
Recurrent Neural Networks (RNN): Elman Architecture

- The Elman architecture is a type of recurrent neural network (RNN) proposed by **Jeffrey Elman in 1990**. It is one of the first and simplest forms of RNN and is based on the idea of using an internal memory to retain information about the past context.
- The Elman architecture consists of three main components: an input layer, a hidden layer (also called the “**state layer**”), and an output layer. These layers are connected to each other through synaptic weights.

Feed Forward Artificial Neural Network (Simple MLP)



Recurrent Neural Network (Elman Architecture)

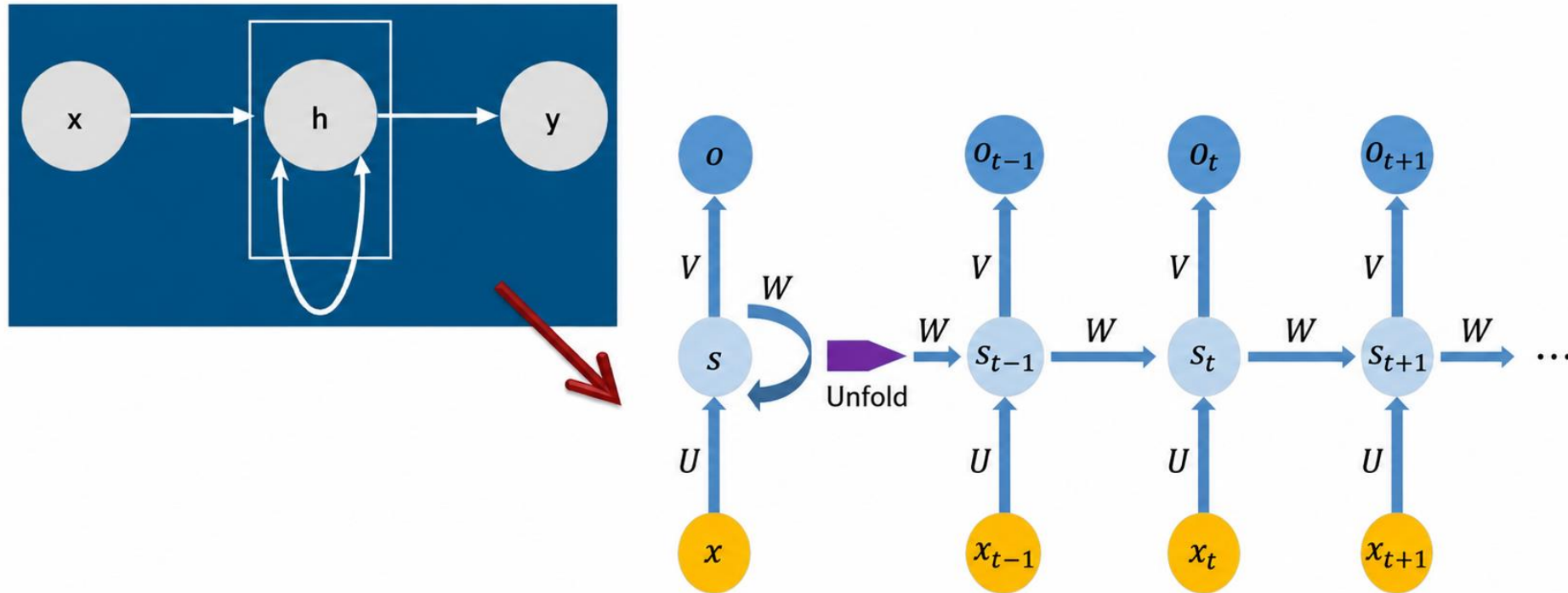


$W_i \quad W_o$

$$h(t) = f(x(t)W_i + h(t-1)W_h + b_h)$$

Unfolding of RNN

- Although RNNs model time series very well, it is difficult to learn long-term dependencies due to the **vanishing gradient** problem in **Back-Propagation Through Time (BPTT)** (Palangi H, et al., 2016).



Back Propagation Through Time (BPTT)

- In BPTT, updating the weights is essentially the same as in normal Back Propagation:
- We can show that the derivative of the “Cross-Entropy” loss function passes through the derivative of the Softmax.

$$p_j = \frac{e^{a_j}}{\sum_{k=1}^K e^{a_k}}$$

- These objects must then be multiplied across their various stages due to the chain rule:

$$d[W_h^T h(t-1)] / dW_h$$

- The result is that the gradients either **vanish** over time (vanishing gradient problem) or become very large quickly (exploding gradient problem).
- **RNNs, GRUs, and LSTMs** solve (or at least mitigate) the vanishing gradient problem present in BPTT.

$$y(t) = \text{softmax}(W_o^T h(t))$$

$$y(t) = \text{softmax}(W_o^T f(W_h^T h(t-1) + W_x^T x(t)))$$

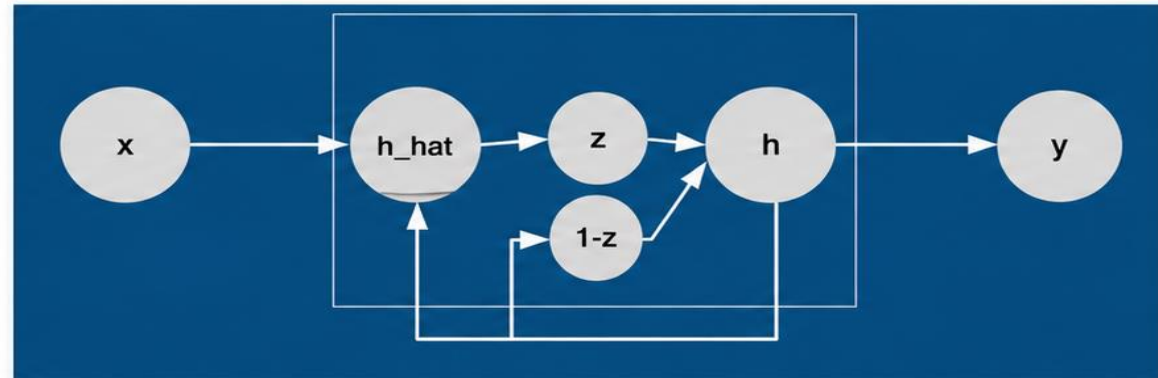
$$y(t) = \text{softmax}(W_o^T f(W_h^T f(W_h^T h(t-2) + W_x^T x(t-1) + W_x^T x(t)))$$

$$y(t) = \text{softmax}(W_o^T f(W_h^T f(W_h^T f(W_h^T h(t-3) + W_x^T x(t-2)) + W_x^T x(t-1) + W_x^T x(t)))$$

- For simplicity, let's remove the bias term from this sequence to make things easier to visualize.

Rated Recurrent Unit (RRU)

- The main idea is to weight $f(x, h(t-1))$, which is the output of a simple RNN, and $h(t-1)$, the previous state (Amari, et al., 1995).
- We add a rating operation between what would be the simple RNN output and the previous state value.
- This new operation can be seen as a gate at a given time that takes values between 0 and 1, while the other internal gate has a value equal to 1 minus the value of zeta.
- This gate chooses between two things: a) keeping the old value or b) taking the new one. As a result, we obtain a mix of the two.



$$h_hat(t) = f(x(t))W_x + h(t-1)W_h + b_h$$

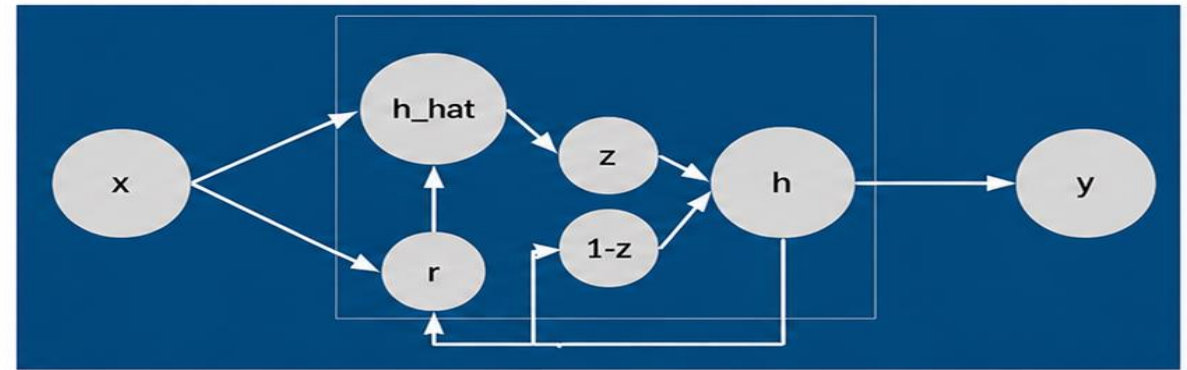
$$z(t) = \text{sigmoid}(x(t)W_{xz} + h(t-1)W_{hz} + b_z)$$

$$h(t) = (1 - z(t)) * h(t-1) + z(t) * h_hat(t)$$

- $z(t)$ is called the “rate”

Gated Recurrent Unit (GRU)

- Gated Recurrent Units (GRU) were introduced in 2014 and are a simpler version of LSTMs. They have fewer parameters but the same concepts (Chung, et al., 2014).
- Recent research has also shown that the accuracy of LSTMs and GRUs is comparable and that GRUs even perform better in some cases.
- In GRUs, we add a gate in addition to those used in RNNs (such as the “reset gate $r(t)$ ”), which controls how much of the previous hidden state will be considered when computing the new candidate hidden value. In other words, this reset gate can completely reset the hidden state.
- The old RNN gate is now called the “update gate $z(t)$ ” and is responsible for balancing the previous hidden values and the new candidate to form a new hidden state.



$$r_t = \sigma(x_t W_{xr} + h_{t-1} W_{hr} + b_r)$$

$$z_t = \sigma(x_t W_{xz} + h_{t-1} W_{hz} + b_z)$$

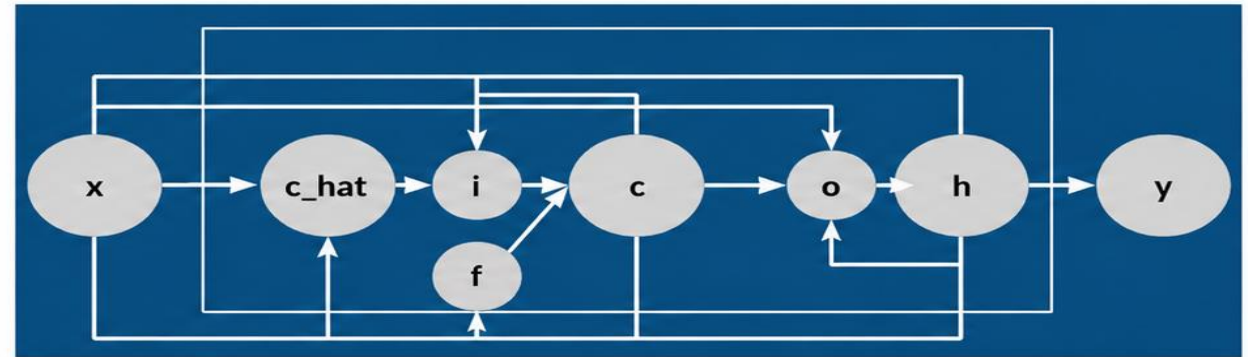
$$\hat{h}_t = g(x_t W_{xh} + (r_t \odot h_{t-1}) W_{hh} + b_h)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \hat{h}_t.$$

Long Short Term Memory (LSTM)

- LSTM is an effective solution to the vanishing gradient problem through the use of memory cells (Hochreiter, et al., 1997).

- A memory cell consists of four units: an input gate, an output gate, a forget gate, and a self-recurrent neuron.
- The gates control the interactions between neighboring memory cells and the memory cell itself.
- If the input signal needs to modify or not the state of the memory cell, this is controlled by the input gate. On the other hand, the output gate can control the state of the memory cell, deciding how much it should or should not affect the state of the adjacent memory cell. In addition, the forget gate can choose to remember or forget the previous state.



$$i_t = \sigma(x_t W_{xi} + h_{t-1} W_{hi} + c_{t-1} W_{ci} + b_i)$$

$$f_t = \sigma(x_t W_{xf} + h_{t-1} W_{hf} + c_{t-1} W_{cf} + b_f)$$

$$c_t = f_t c_{t-1} + i_t \tanh(x_t W_{xc} + h_{t-1} W_{hc} + b_c)$$

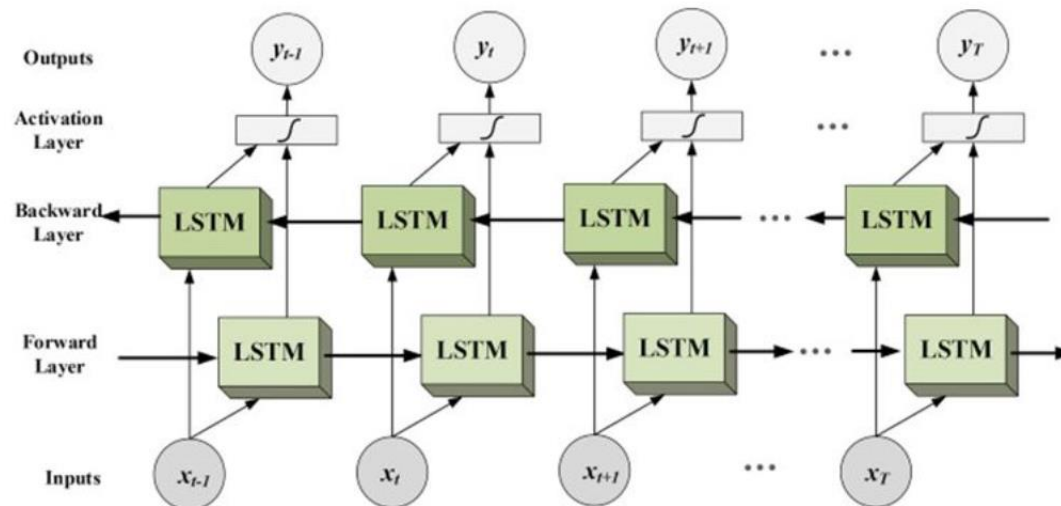
$$o_t = \sigma(x_t W_{xo} + h_{t-1} W_{ho} + c_t W_{co} + b_o)$$

$$h_t = o_t \tanh(c_t)$$

Bidirectional LSTM

We use a **Bidirectional** container for recurrent neural networks (RNNs), and these two Bidirectional LSTM layers are useful for modeling the order of words in a sequence in both directions.

Bidirectional Long Short-Term Memory networks (Bi-LSTMs) make any neural network capable of handling sequential information in both directions: “backward” (from future to past) and “forward” (from past to future). In a bidirectional architecture, the input flows in two directions, which makes a Bi-LSTM different from a standard LSTM. With a regular LSTM, the input can flow in only one direction. In contrast, in a bidirectional model, the inputs can flow in both directions, allowing the network to preserve both past and future information.



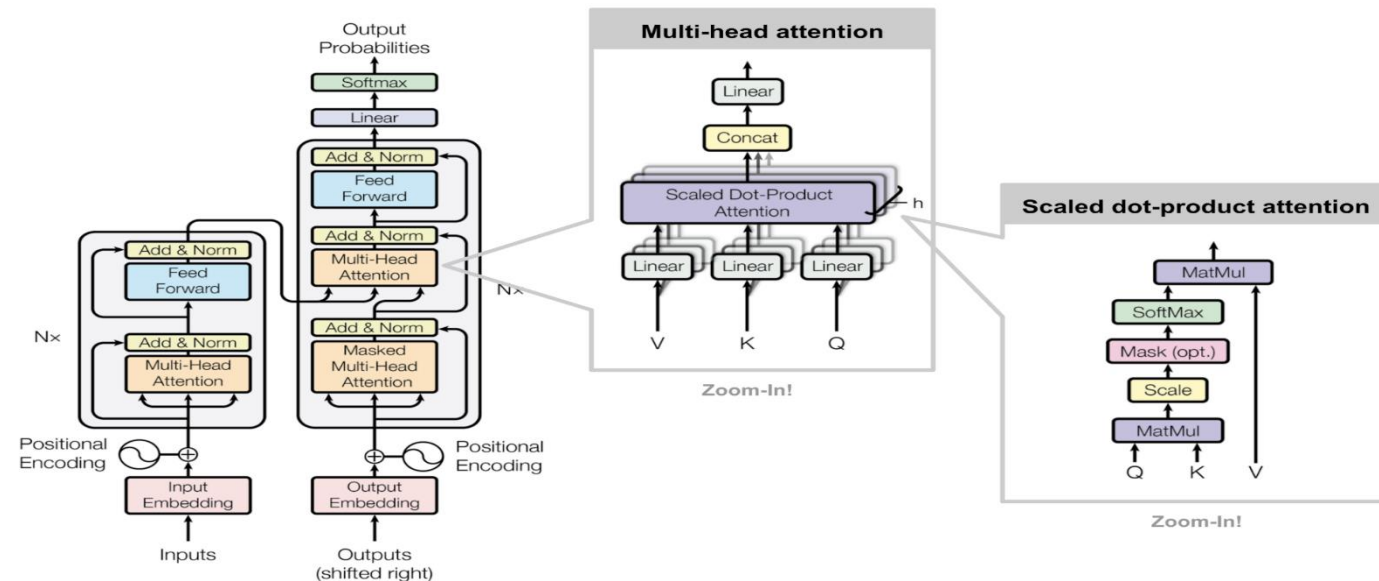
The Innovation of Transformers

- ✓ **Transformers** change all the game in **Natural Language Processing (NLP)**, which has become one of the pillars of artificial intelligence in a global digital economy.
- ✓ **Transformer** models are disruptive and are leading to a new age in **Artificial Intelligence**. Language understanding has become the mile-stone of language modeling, chatbots, personal assistants, question answering, text summarizing, speech-to-text, sentiment analysis, machine translation, and more.
- ✓ By means of **Transformers**, **Artificial Intelligence** has evolved to billion-parameter models in order to address the challenge of trillion-word datasets.
- ✓ They get far from the past, leaving the dominance of **RNNs** and **CNNs** behind. **BERT** and **GPT** models leave the recurrent network layers back and replace them with self-attention mechanisms.



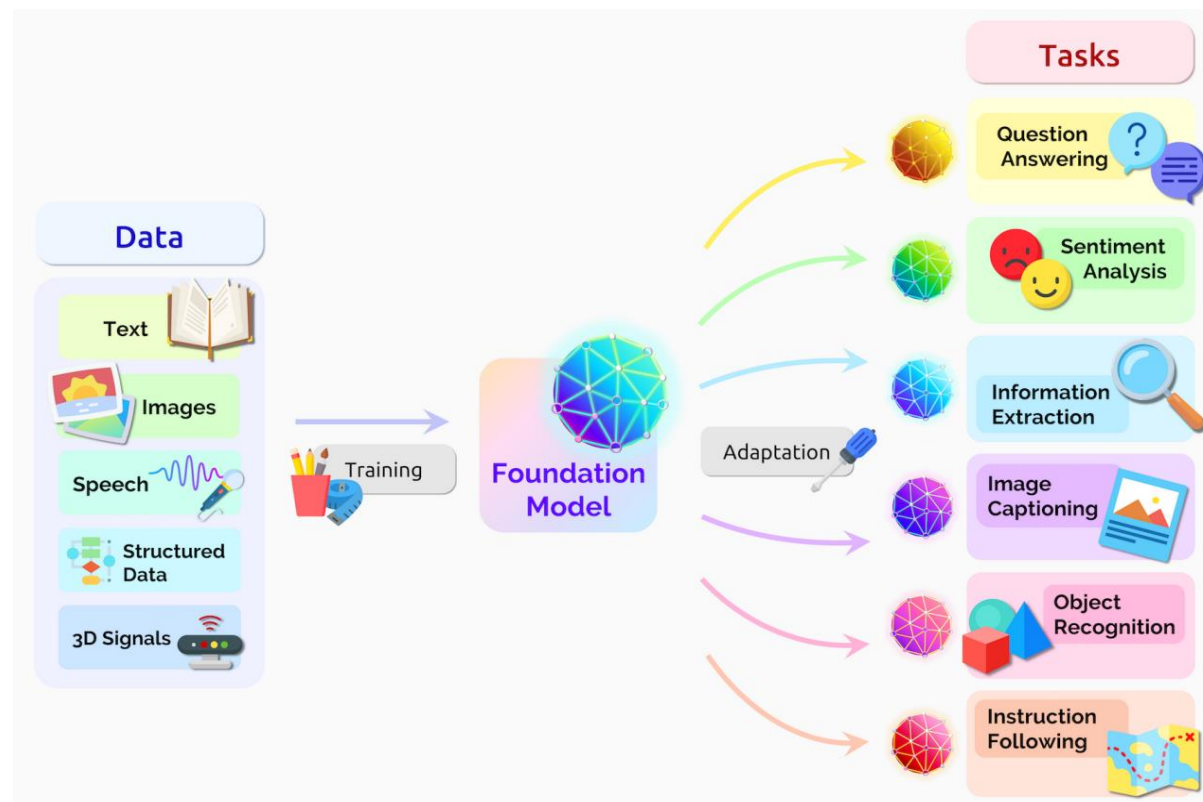
The increasing amount of Data Dilemma

- ✓ The increasing amount of data requires training AI models at a very large scale, therefore **Transformers** launch new era of **parameter-driven AI**. Learning to understand how hundreds of millions of words fit together in sentences requires a tremendous amount of parameters.
- ✓ **Transformers encoders** and **decoders** contain **attention heads** that train separately, parallelizing cutting-edge hardware. Attention heads can run on separate **GPUs** opening the door to billion-parameter models and soon-to-come trillion-parameter models. **OpenAI** trained a 175 billion parameter **GPT-3** Transformer model on a supercomputer with **10,000 GPUs** and **285,000 CPU** cores.



What are Foundation Models?

- ✓ **Foundation Models** are fully trained transformer models that can perform hundreds of tasks even without fine-tuning. Foundation models at this scale offer the tools we need in this massive information era.
- ✓ In the field of **Sentiment Analysis** for example, how many humans it would take to read the content of billions of tweets or posts on social networks in a few minutes?
- ✓ How many humans would be needed to translate the millions of **pages published** each day on the web?



- ✓ How many humans it would take to write the transcripts of all of the vast amount of hours of **streaming published** per day on the web?
- ✓ How many humans are required to replace **AI image captioning** for the billions of images that continuously appear online?

AI 2.0 and Industry 4.0

- ✓ **Transformers** are **industrialized, homogenized** deep learning models designed for **parallel** computing on **supercomputers**.
- ✓ Thanks to **homogenization**, one transformer model addresses a wide range of tasks with no fine-tuning. Transformers can perform self-supervised learning on billions of records of raw unlabeled data with billions of parameters.
- ✓ These particular architectures are called foundation models. **Foundation model** transformers represent the edge of the **Fourth Industrial Revolution** that began in **2015** with machine-to-machine automation that will connect everything to everything.
- ✓ **Artificial intelligence** in general and specifically **Natural Language Processing (NLP)** for **Industry 4.0 (I4.0)** has gone far beyond the software practices of the past.



- ✓ An **Industry 4.0** Project Manager may go to OpenAI's cloud platform, sign up, obtain an **API key**, and get to work in a few minutes. On the other side, a user can enter a text, specify the NLP task, and obtain a response sent by a GPT-3 transformer.
- ✓ Finally, a user can go to **GPT-3 Codex** and develop programs without any coding knowledge.

AI 2.0 and Industry 4.0

- ✓ A GPT-3 model might not fit a specific task. For example, a project manager, consultant, or developer might want to use another system provided by Google AI, Amazon Web Services (AWS), the Allen Institute for AI, or Hugging Face.
- ✓ Should a Project Manager choose to work locally? Or should the implementation be done directly on Google Cloud, Microsoft Azure, or AWS? Yes
- ✓ Should a development team select Hugging Face, Google Trax, OpenAI, or AllenNLP? Yes
- ✓ Should an **Artificial Intelligence** specialist or a **Data Scientist** use an API with practically no AI development? Yes

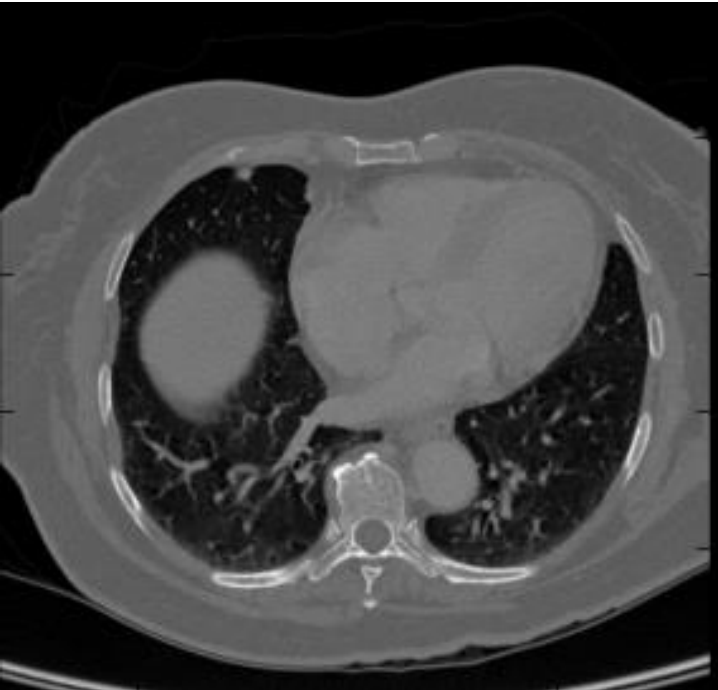


HUGGING FACE



Use Case 1: AI for Lung Cancer Detection

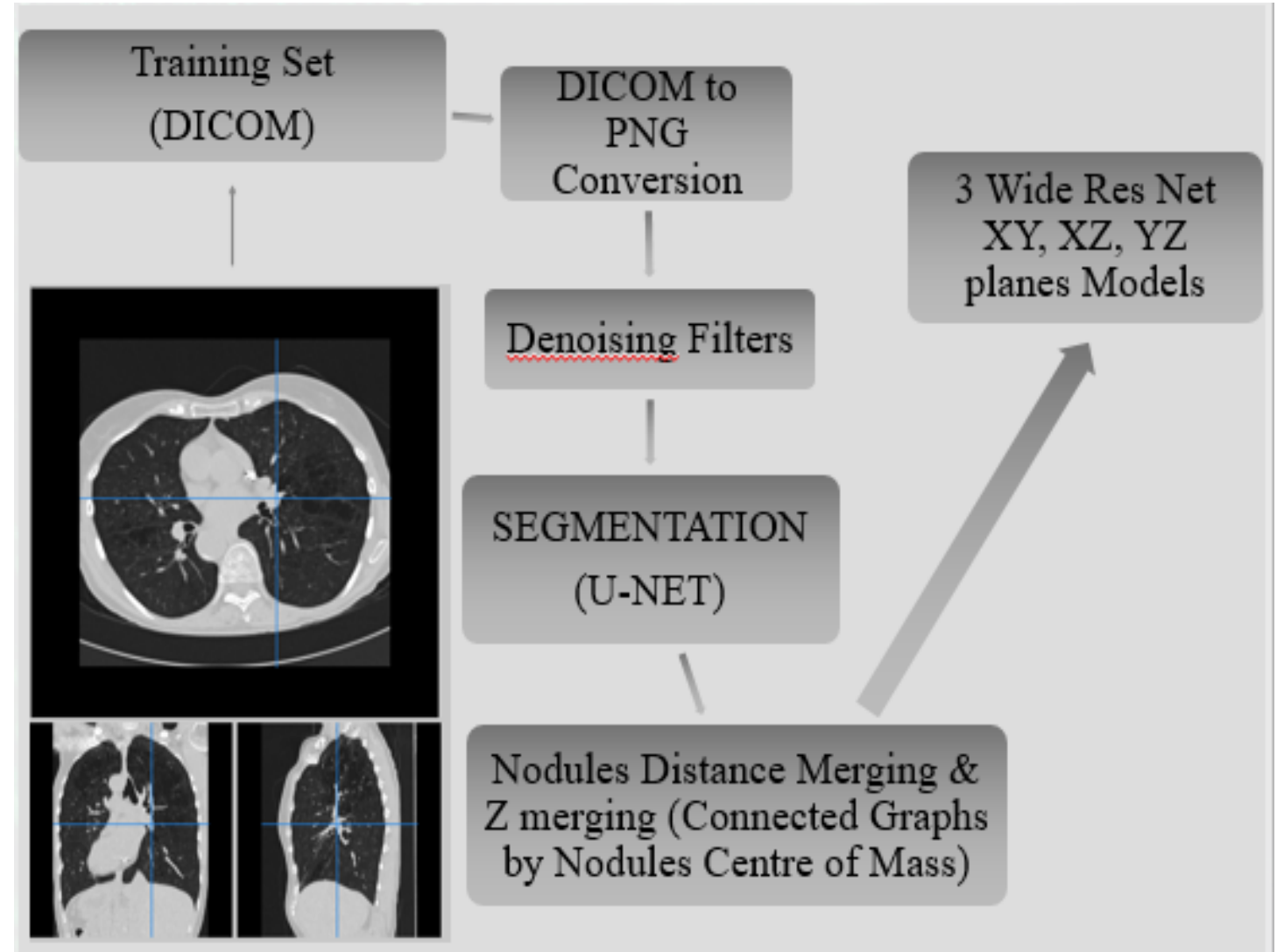
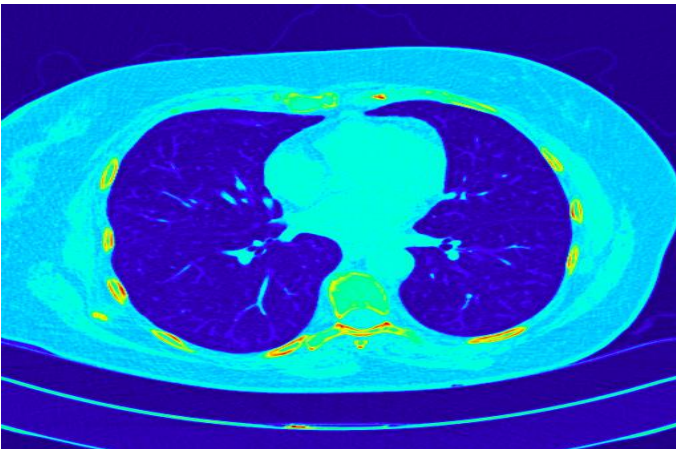
Lung Cancer Dataset



- **Kaggle:** In 2010, Kaggle was founded as a platform for predictive **modeling** and **analytics competitions** on which companies and researchers post their data.
- Statisticians and data scientists from all over the world compete to produce the best models.
- **Data Science Bowl 2017** was the biggest competition focused on “Lung Cancer Detection”. The competition was founded by **Arnold Foundation** and awarded **\$1 million in prizes** (1st ranked \$500,000).
- **Train Set:** around 150 CT labelled scans images per patient from 1200 patients encoded in **DICOM** format.
- **Stage 1 test set:** 190 patients CT scans.
- **Stage 2 test :** 500 patients CT scans.

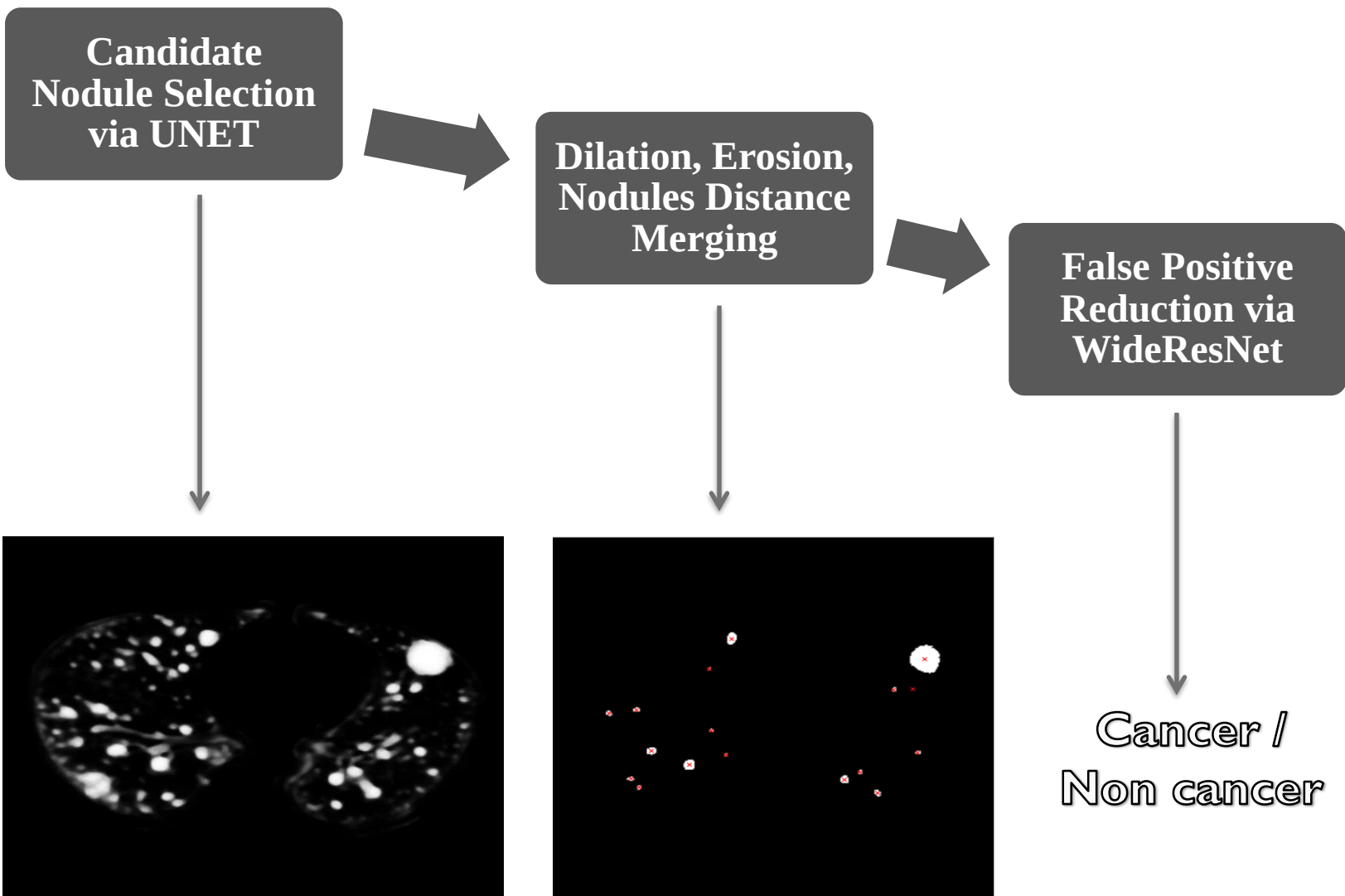
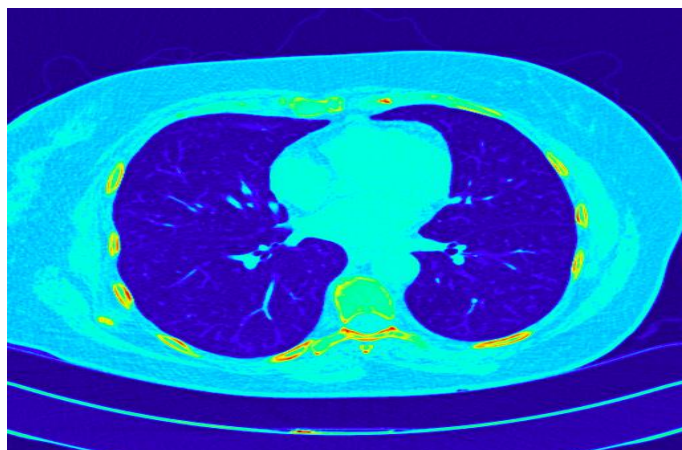
Use Case 1: AI for Lung Cancer Detection

Lung Cancer Models



Use Case 1: AI for Lung Cancer Detection

Lung Cancer Results



Use Case 2: AI for the automatic classification of Causes of Death

Detection Model

For each individual death occurring within the national territory, the certifying physician records the information on the causes of death in this part of the paper certificate

ATTENZIONE
leggere le ISTRUZIONI SUL RETRO
scrivere in STAMPATELLO

Istat
ISTITUTO NAZIONALE DI STATISTICA
SCHEDA DI MORTE OLTRE IL 1° ANNO DI VITA

ISTAT MODELLO D4 - Ediz. 2017
COPIA PER LISTAT

PARTE A - A CURA DEL MEDICO
Indicare la causa di morte e le condizioni morbose o traumatiche che hanno contribuito al decesso.

1. Luogo del decesso
Indicare il luogo in cui è avvenuto il decesso.
Abitazione _____ Comune _____
Altrove _____
Indicare il tipo di struttura ospedaliera o sanitaria (es. ospedale, clinica, ecc.) _____
Indicare il tipo di struttura (es. ospedale, clinica, ecc.) _____

2. Ricovero diagnostico
Indicare il ricovero diagnostico.
Sì _____ No _____
Indicare il tipo di ricovero (es. ricovero ospedaliero, ecc.) _____

3. Cause iniziali
Indicare la causa iniziale che ha dato inizio alla sequenza.
Sì _____ No _____
Indicare il tipo di causa (es. trauma, malattia, ecc.) _____

4. Parte I
CAUSA DI MORTE: sequenza di condizioni morbose o traumatiche o avvelenamenti che ha condotto a morte - In presenza di più sequenze scegliere la più rilevante.

5. Data di morte
Indicare la data di morte.
Giorno _____ Mese _____ Anno _____

6. Località di nascita
Indicare la località di nascita.
Comune _____
Indicare il tipo di località (es. città, paese, ecc.) _____

7. Data di nascita
Indicare la data di nascita.
Giorno _____ Mese _____ Anno _____

8. Sesso
Indicare il sesso.
Maschio _____ Femmina _____

9. Stato civile
Indicare lo stato civile.
Sposato _____ Vedovo _____ Divorziato _____ Single _____

10. Grado di istruzione
Indicare il grado di istruzione.
Scuola elementare _____ Scuola media _____ Scuola superiore _____ Università _____

11. Luogo di nascita
Indicare il luogo di nascita.
Comune _____
Indicare il tipo di località (es. città, paese, ecc.) _____

12. Luogo di residenza
Indicare il luogo di residenza.
Comune _____
Indicare il tipo di località (es. città, paese, ecc.) _____

13. Data dell'incidente, infortunio, suicidio, omicidio
Indicare la data dell'incidente, infortunio, suicidio, omicidio.
Giorno _____ Mese _____ Anno _____

14. Modalità del trauma/avvelenamento
Indicare la modalità del trauma/avvelenamento.
Accidentale _____ Suicidio _____ Omicidio _____

15. Infortunio sul lavoro
Indicare se si tratta di un infortunio sul lavoro.
Sì _____ No _____

16. Mezzo di trasporto
Indicare il mezzo di trasporto.
Auto _____ Moto _____ Altro _____

17. Ruolo della vittima
Indicare il ruolo della vittima.
Passeggero _____ Conduttore _____ Altro _____

18. Tipo di incidente
Indicare il tipo di incidente.
Incendio _____ Esplosione _____ Altro _____

19. Tipo di investimento
Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

20. Tipo di investimento
Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

21. Tipo di investimento
Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
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Indicare il tipo di investimento.
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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

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Investimento in terra _____ Investimento in mare _____ Altro _____

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Investimento in terra _____ Investimento in mare _____ Altro _____

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Investimento in terra _____ Investimento in mare _____ Altro _____

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Investimento in terra _____ Investimento in mare _____ Altro _____

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Investimento in terra _____ Investimento in mare _____ Altro _____

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Investimento in terra _____ Investimento in mare _____ Altro _____

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Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

100. Tipo di investimento
Indicare il tipo di investimento.
Investimento in terra _____ Investimento in mare _____ Altro _____

Istat Form D4-D4bis 'Death Certificate

4. Parte I
CAUSA DI MORTE: sequenza di condizioni morbose o traumatiche o avvelenamenti che ha condotto a morte - In presenza di più sequenze scegliere la più rilevante.

Causa iniziale. Scegliere la SOLA patologia o traumatismo che ha dato inizio alla sequenza.

1. che ha causato

2. che ha causato

3. che ha causato

4. che ha causato

Tempo intercorso tra l'insorgenza della causa indicata e la morte

anni o mesi o giorni

anni o mesi o giorni

anni o mesi o giorni

anni o mesi o giorni

4. Parte II
Altri stati morbosi rilevanti - Indicare altre malattie o condizioni morbose o traumatiche esclusi dalla sequenza riportata nel quesito 4. Parte I, ma che hanno contribuito al decesso.

anni o mesi o giorni

anni o mesi o giorni

anni o mesi o giorni

N.B. I QUESITI 5-9 DEVONO ESSERE COMPILATI SOLO IN CASO DI MORTE DA TRAUMATISMO/AVVELENAMENTO (anche con informazioni riferite)

5. Circonstanza che ha dato origine al trauma/avvelenamento indicato in 4. Parte I (es. caduta da scala a pioli, impiccamento, fucilata, ecc.)

6. Modalità del trauma/avvelenamento

Accidentale (include incidenti da trasporto)....1 ☐ Suicidio....2 ☐ Omicidio....3 ☐

6.1 Infortunio sul lavoro 1 ☐ 2 ☐

8. Data dell'accidente, infortunio, suicidio, omicidio

Orario _____ giorno _____ mese _____ anno _____

Use Case 2: AI for the automatic classification of Causes of Death

The project aims to:

- identify the most suitable AI methods for coding causes of death;
- create an algorithm for coding causes of death by applying these methods;
- evaluate the performance of automatic coding and the quality of the data by applying this algorithm.

The main expected results are:

- a reduction in coding variability among coders and in errors, resulting in improved quality;
- in view of the imminent replacement of the paper form with an electronic model, AI methods will be useful to maintain high performance in recognizing medical text, even in the absence of the preprocessing currently applied during the registration of paper forms.
- an increase in the number of death certificates coded completely automatically, without human intervention;

Use Case 2: AI for the automatic classification of Causes of Death

The information on the causes is recorded by physicians on a paper death certificate, digitized, and coded according to the **tenth revision of the International Classification of Diseases (ICD-10)** of the **World Health Organization (WHO)**.

The application of the ICD is a complex process that involves:

- first, assigning a code to all the causes certified by physicians on the forms (multiple causes).
- then, selecting only one code for each death: the **underlying cause**.

Both phases are carried out according to the rules described in the ICD, and both the multiple causes and the underlying cause are used for mortality statistics.

Use Case 2: AI for the automatic classification of Causes of Death

Training set 22_23: about 4 million texts.

53,876 classes after class cleaning and drops with cutoff.

Too much data for training: we sample progressively (starting from 300,000 texts).

DS_Gold_standard (external): 951 texts assigned a code whose correctness has been verified.

TEST_08_09 (test_set_blind): 11,909 texts to be labeled.

Valid ICD codes

Flag (A, C, CON, P, etc). Example:

Separators (;, /, etc). Example: *CACHESSIA TERMINALE CON IRA IN GRANDE ANZIANO* coded as **NI79 / R54**.

(Note: “*Cachessia terminale con IRA in grande anziano*” = “*Terminal cachexia with acute renal failure in a very elderly patient.*”)

Use Case 2: AI for the automatic classification of Causes of Death

TF-IDF on the dataset to obtain the normalized term-document matrix; **Random Forest** is used for classification (which works on SL).

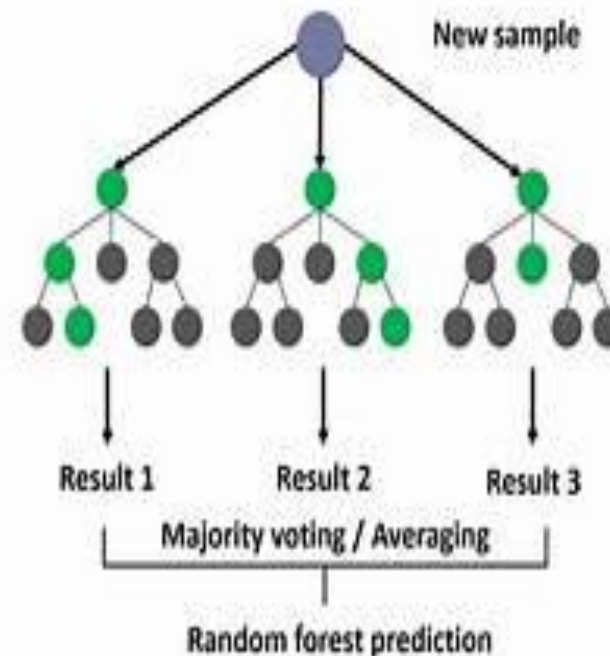
We take a sample of **300,000 texts** for training.

Initially **54,767 classes**, later reduced to **27,364**, after non-semantic splitting (in cases where the number of classes equals the number of texts).

We use a **cutoff of 2**, meaning we consider only classes with at least two instances.

There are about **17,666 classes** that appear only once, and these are removed.

In the end, we are left with **9,698 classes**.



Use Case 2: AI for the automatic classification of Causes of Death

On training set of 300,000:

Accuracy: 0.9761%

F1-score (macro-weighted): 0.9745%

On test set (20%):

Accuracy: 0.9467%

F1-score (macro-weighted): 0.9422%

Accuracy on test set of about 65,161 texts:

- #codes = #texts (size = 63,636, 97.66% of the test dataset): Accuracy: 0.9528% – F1-score (macro-weighted): 0.9505%
- #codes > #texts (size = 1,525, 2.34% of the test dataset): Accuracy: 0.69% – F1-score (macro-weighted): 0.71% → 1,060 correct, 465 incorrect (outputs exported to Excel and CSV for sharing)
- #texts > #codes (size = 0)

(Note: The “%” signs seem to be used instead of decimals — e.g., accuracy 0.9761 likely means 97.61%.)

Use Case 2: AI for the automatic classification of Causes of Death

Accuracy on test set of about 65,161 texts:

#codes > #texts (size = 1,525, 2.34% of the test dataset): Accuracy: 0.69% – F1-score (macro-weighted): 0.71% → 1,060 correct, 465 incorrect.

The correct cases are mostly those in which one text corresponds to two codes.

We check what percentage of the texts corresponding to the correct cases are present in the training set: **915 out of 1,050 (86%)**.

We check what percentage of the texts corresponding to the incorrect cases are present in the training set: **76 out of 475 (16%)**.

Theory: Random Forest performs well when it finds the same words combined in the same way in the training set.

Use Case 2: AI for the automatic classification of Causes of Death

Accuracy on test set (sample of 950 “gold_standard” texts):

For this dataset, the initial number of classes is **876**. After non-semantic splitting (in cases where the number of classes equals the number of texts — i.e., for 197 cases), we obtain **847 classes**, and the number of texts to be classified becomes **1,154**.

The classes are reduced to **716** after removing the time period.

Accuracy on “gold_standard” test set: overall **0.41% F1-score** → problem: only **24% of the gold standard texts** are present in the training set, while **62% of the classes** are present in the training set.

- #codes = #texts (size = 948, 82.15% of the gold standard dataset): Accuracy: 0.55% – F1-score (macro-weighted): 0.52%

- #codes > #texts (size = 206, 17.85% of the gold standard dataset): Accuracy: 0.033% – F1-score (macro-weighted): 0.033% → 7 correct, 199 incorrect (outputs exported to Excel and CSV for sharing)

- #texts > #codes (size = 0)

OBSERVATION: for the label encoding of the classes, we include both the classes present in the training set and those in the gold standard, to prevent cases where classes in the gold standard are missing from the training data.

In the previous case, this issue did not arise because we had obtained it using a **stratified train-test split**, which ensures that each class has a sample in both the training and test sets.

Use Case 3: Forecasting the number of Intensive Care Beds occupied by COVID-19 patients through the use of Recurrent Neural Networks

ICU Forecasting



AI FOR ICU FORECASTING

Better predictions, better decisions, better care.

- Since **2019**, **COVID-19** has caused millions of deaths worldwide and a major global economic crisis.
- Accurate forecasting of **ICU (Intensive Care Beds)** bed occupancy can improve hospital resource **allocation** and **optimize** public healthcare spending.
- Early **ICU forecasts** can help governments introduce targeted containment measures and mobility restrictions sooner.
- Anticipating ICU overload may reduce the duration and severity of lockdown measures.
- The study proposes a fully automated COVID-19 forecasting framework based on **Deep Learning** techniques.

Use Case 3: Forecasting the number of Intensive Care Beds occupied by COVID-19 patients through the use of Recurrent Neural Networks

- The models used include **LSTM** and **Bidirectional LSTM** (Bi-LSTM) architectures enhanced with “Inference Dropout” regularization.
- Results show **state-of-the-art** accuracy in ICU occupancy prediction, making the system a valuable tool for policymakers and health-care planning.
- Among the tested architectures, **Bi-LSTM** achieved the most stable forecasting performance, outperforming standard **LSTM**.

Value of metrics for the ARIMA model					
	Period 1	Period 2	Period 3	Period 4	Period 5
MAE	10.123	5.433	6.487	13.466	3.286
RMSE	11.905	7.039	7.628	15.288	3.679

Table 7 Value of metrics for the Bi-LSTM ensemble model					
	Period 1	Period 2	Period 3	Period 4	Period 5
MAE	6.547	14.391	18.284	12.407	7.799
RMSE	8.381	19.058	19.283	12.645	8.402

Table 8 Value of metrics for the LSTM ensemble model					
	Period 1	Period 2	Period 3	Period 4	Period 5
MAE	5.8	15.007	13.369	16.48	7.831
RMSE	7.825	18.946	14.352	16.572	8.332

Table 9 Value of metrics for the Bi-LSTM model with non-shifted data					
	Period 1	Period 2	Period 3	Period 4	Period 5
MAE	15.482	33.996	23.941	5.097	8.69
RMSE	18.659	36.208	24.974	5.659	9.737

Table 10 Value of metrics for the LSTM model with non-shifted data					
	Period 1	Period 2	Period 3	Period 4	Period 5
MAE	21.67	36.227	35.01	4.914	4.118
RMSE	23.453	38.332	35.522	5.462	5.084

Use Case 3: Forecasting the number of Intensive Care Beds occupied by COVID-19 patients through the use of Recurrent Neural Networks

- The framework could be **extended** to predict **additional healthcare indicators**, such as total hospitalized patients.
- The model may **support** the Italian **regional risk** classification system by estimating the **probability** of moving between risk zones.
- As more data become available over time, the **forecasting system** is expected to become increasingly accurate and effective.

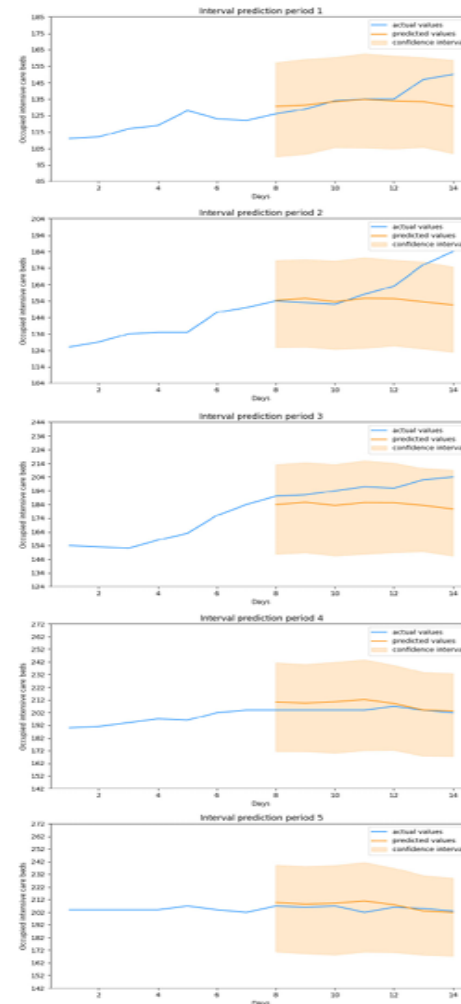


Fig. 1. Bi-LSTM predictions.

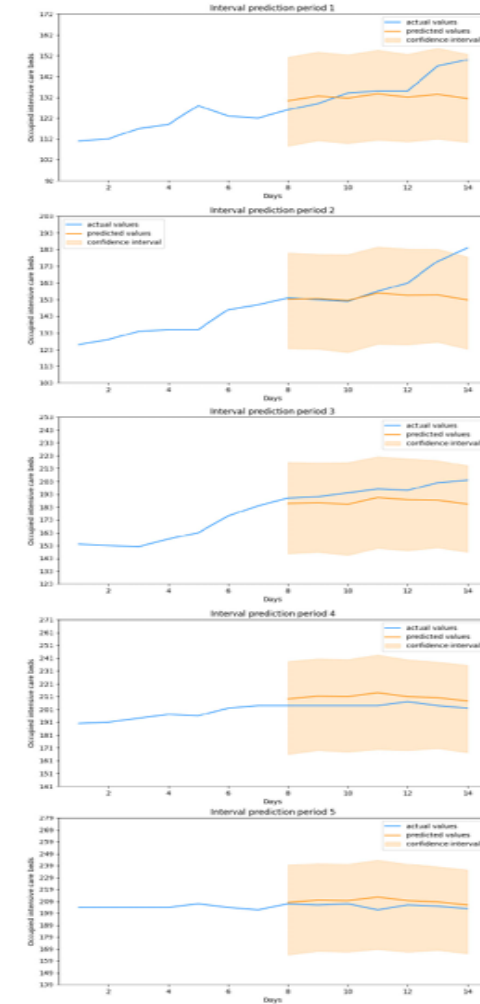


Fig. 2. LSTM predictions.

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Thanks

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